

To chaliye shuru karte hain ek scoring aur mazedaar chapter, jisko hum chaar parts mein cover karenge — (thermal expansion, calorimetry, conduction, radiation).

JEE Main / NEET ka yeh favourite part hai, aur jitna content yahan diya ja raha hai, agar use utna achhe se kar liya, to aap Advance PYQs ke lagbhag 80% cover kar loge.

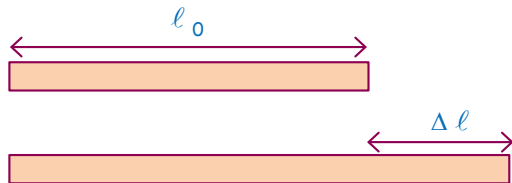
Let's start.



THERMAL EXPANSION

Consider a rod of length ℓ_0 at temp θ_i .

If temperature is increase by $\Delta\theta$ or ΔT .



Experimentally

$$\Delta \ell \propto \Delta \theta$$

$$\propto \ell_0$$

$$\Delta \ell = \alpha \ell_0 \Delta \theta$$

$$\Delta \ell = \ell_f - \ell_0 = \ell_0 \alpha \Delta \theta$$

$$\ell_f = \ell_0 + \ell_0 \alpha \Delta \theta$$

$$\ell_f = \ell_0(1 + \alpha \Delta \theta) \quad \Delta \theta = \theta_f - \theta_i$$

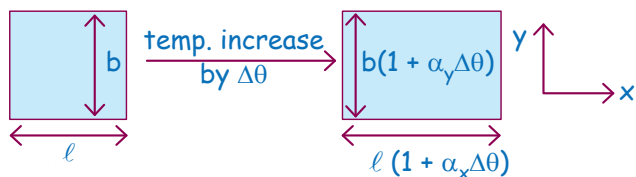
Here ℓ_0 = initial length of rod at temp θ_i

ℓ_f = final length of rod at temp θ_f

α = Linear coefficient of thermal expansion

Areal Expansion Superficial Expansion

Consider we have rectangle of length ℓ and width b at temp θ_i . Now if we increase the temp by $\Delta\theta$ then



$$A_{\text{initial}} = A_0 = \ell b$$

$$A_{\text{final}} = \ell(1 + \alpha_x \Delta \theta) b(1 + \alpha_y \Delta \theta)$$

$$A_f = \ell b (1 + \alpha_y \Delta \theta + \alpha_x \Delta \theta + \alpha_x \alpha_y (\Delta \theta)^2)$$

$$A_f = A_i (1 + (\alpha_x + \alpha_y) \Delta \theta)$$

Neglect

$$A_f = A_0 (1 + \beta \Delta \theta)$$

Where $\beta = \alpha_x + \alpha_y$

Similarly we can do same calculation for volume expansion.

Le bhai, ab purani baatein bhool jao, aur maine neeche saare formula badhiya se likh diye hain. Practice them on rough paper.

Linear Expansion

$$\star \Delta \ell = \ell_0 \alpha \Delta \theta$$

$$\star \frac{\Delta \ell}{\ell_0} = \alpha \Delta \theta$$

$$\star \ell_f = \ell_0 (1 + \alpha \Delta \theta)$$

Areal Expansion

$$\star \Delta A = A_0 \beta \Delta \theta$$

$$\star \frac{\Delta A}{A_0} = \beta \Delta \theta$$

$$\star A_f = A_0 (1 + \beta \Delta \theta)$$

$$\beta = \alpha_x + \alpha_y$$

Volume Expansion

$$\star \Delta V = V_0 \gamma \Delta \theta$$

$$\star \frac{\Delta V}{V_0} = \gamma \Delta \theta$$

$$\star V_f = V_0 (1 + \gamma \Delta \theta)$$

$$\star \gamma = \alpha_x + \alpha_y + \alpha_z$$

Isotropic Material

$$\alpha_x = \alpha_y = \alpha_z = \alpha \text{ (let)}$$

$$\Delta l = l_0 \alpha \Delta \theta$$

$$\Delta A = A_0 \beta \Delta \theta = A_0 (2\alpha) \Delta \theta$$

$$\Delta V = A_0 \gamma \Delta \theta = V_0 (3\alpha) \Delta \theta$$

$$\beta = \alpha_x + \alpha_y = 2\alpha$$

$$\gamma = \alpha_x + \alpha_y + \alpha_z = 3\alpha$$

$$\alpha : \beta : \gamma \Rightarrow 1 : 2 : 3$$

Variation of Density on Changing Temperature

If we ↑ the temperature by ΔT

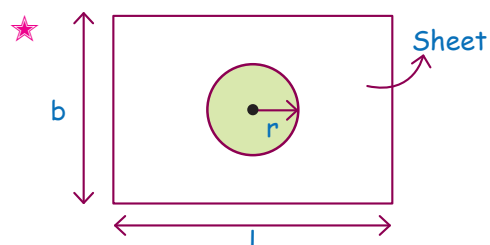
$$\rho_0 = \frac{m}{V_0}$$

$$\rho_{\text{new}} = \frac{m}{V_0(1 + \gamma\Delta T)}$$

$$\rho_{\text{new}} = \rho_0(1 + \gamma\Delta T)^{-1} \text{ Binomial approx.}$$

$$\rho_{\text{new}} = \rho_0(1 + \gamma\Delta T)^{-1}$$

$$\rho = \frac{\rho_0}{1 + \gamma\Delta T} = \rho_0(1 - \gamma\Delta T)$$



If temperature is ↑ (isotropically)

⇒ $l \uparrow, b \uparrow, r \uparrow, x \uparrow$

Q. Consider two rod (L_1, α_1) and (L_2, α_2). If temperature of the rod is increase by $\Delta T^\circ\text{C}$. Find the condition so that

1. Difference of the rod does not change.
2. Combination of rod does not change.

Sol. 1. $\Delta l_1 = \Delta l_2$

$$l_1\alpha_1\Delta T = l_2\alpha_2\Delta T$$

$$l_1\alpha_1 = l_2\alpha_2$$

2. $(L_1 + L_2)_{\text{initial}} = (L_1 + L_2)_{\text{final}}$

$$L_1 + L_2 = L_1(1 + \alpha_1\Delta T) + L_2(1 + \alpha_2\Delta T)$$

$$L_1 + L_2 = L_1 + L_1\alpha_1\Delta T + L_2 + L_2\alpha_2\Delta T$$

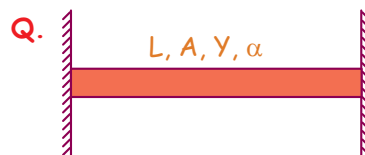
$$L_1\alpha_1 + L_2\alpha_2 = 0$$

$$L_1\alpha_1 = -L_2\alpha_2$$

or method 2:

$$\Delta l_1 = -\Delta l_2$$

$$l_1\alpha_1\Delta T = -l_2\alpha_2\Delta T$$



If temperature of the rod is increase by $\Delta\theta$ (Bending is not allow). Find

1. force applied by wall on rod.
2. find stress and strain develop.

Sol.



Since length of the rod is constant so, we can say rise in temperature will try to expand the rod by Δl_1 and normal from both the wall will try to compress the rod by Δl_2 . So,

$$\Delta l_1 = \Delta l_2$$

$$l\alpha\Delta\theta = \frac{Fl}{Ay}$$

$$F = Ay\alpha\Delta\theta = N_{\text{wall}}$$

$$\frac{F}{A} = \text{Stress} = y\alpha\Delta\theta$$

$$\frac{F}{A} = y\frac{\Delta l}{l}$$

$$F = N_{\text{wall}}$$

$$\text{Strain} = \frac{\Delta l}{l} = \alpha\Delta\theta$$

Upar wale question ki achhe se practice kar lena, kyunki yeh exam mein bahut baar aa chuka hai.

Yahi question rod ki jagah string banakar, temperature decrease karke bhi poocha ja sakta hai.

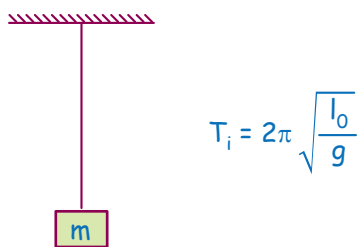


Q. A wire of cross-sectional area 3mm^2 is first stretched between two fixed points at a temperature of 20°C . Determine the tension when the temperature falls to 10°C . Coefficient of linear expansion $\alpha = 10^{-5}^\circ\text{C}^{-1}$ and $Y = 2 \times 10^{11} \text{ N/m}^2$.

Sol. $F = YA \alpha \Delta t$

$$= 2 \times 10^{11} \times 3 \times 10^{-6} \times 10^{-5} \times (20 - 10) = 60 \text{ N}$$

SIMPLE PENDULUM



If temperature is increased by $\Delta\theta$

Temperature $\uparrow \Rightarrow \ell \uparrow, T \uparrow \Rightarrow$ Slow

$$T_f = 2\pi \sqrt{\frac{\ell_f}{g}} = \sqrt{\frac{\ell_0(1 + \alpha\Delta\theta)}{g}}$$

$$T_f = 2\pi \sqrt{\frac{\ell_0}{g}} = \sqrt{1 + \alpha\Delta\theta}$$

$$T_f = T_i(1 + \alpha\Delta\theta)^{1/2} \quad \text{Binomial theorem}$$

$$T_f = T_i + \frac{T_i\alpha\Delta\theta}{2}$$

$$\frac{\Delta T}{T_i} = \frac{T_f - T_i}{T_i} = \frac{\alpha\Delta\theta}{2} \quad \text{Time loss per second}$$

Method 2:

$$T = 2\pi \sqrt{\frac{I}{g}} = \sqrt{\frac{2\pi}{g}} I^{1/2}$$

$$\frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta I}{I} = \frac{1}{2} \alpha\Delta\theta$$

Use this: Ye cheej hum error mein padhenge.

If $y = ka^2b^3$ (k is constant)

$$\frac{\Delta y}{y} = 2\frac{\Delta a}{a} + 3\frac{\Delta b}{b}$$



Q. Estimate the time lost or gained by a pendulum clock at the end of a week when the atmospheric temperature rises to 40°C . The clock is known to give correct time at 15°C and the pendulum is of steel. (Coefficient of linear expansion of steel is $12 \times 10^{-6} / ^\circ\text{C}$).

Sol. Rise in temperature = $40 - 15 = 25^\circ\text{C}$

$$T = 2\pi \sqrt{\frac{\ell}{g}}$$

$$\frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta \ell}{\ell}$$

$$\text{Time lost per second} = \frac{1}{2} \alpha\Delta\theta$$

$$= \frac{1}{2} \times (12 \times 10^{-6} / ^\circ\text{C}) \times (25^\circ\text{C}) = 150 \times 10^{-6} \text{ s/s}$$

$$\text{Therefore, time lost per week (i.e., } 7 \times 86400 \text{ s)} \\ = 150 \times 10^{-6} \times 7 \times 86400 \text{ s} = 90.72 \text{ sec.}$$

Q. If temperature is increase by 60°C find % increase in moment of inertia. ($\alpha = 10^{-5} / ^\circ\text{C}$)



$$\text{Sol. } I = \frac{m}{3} l^2$$

$$\frac{\Delta I}{I} = 2 \frac{\Delta l}{l} = 2(\alpha\Delta\theta)$$

$$y = 5a^2b^3c$$

$$\frac{\Delta y}{y} = 2\frac{\Delta a}{a} + 3\frac{\Delta b}{b} + \frac{\Delta c}{c}$$

$$\frac{\Delta I}{I} = 2 \times 10^{-5} \times 60 = 120 \times 10^{-5}$$

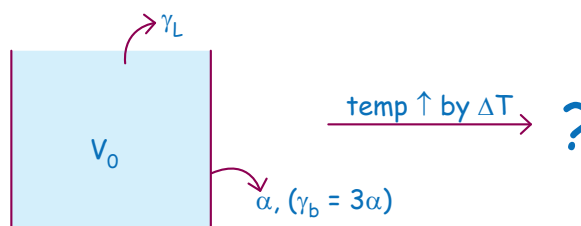
$$\frac{\Delta I}{I} \times 100 = \% \text{ change in}$$

$$= 120 \times 10^{-5} \times 100$$

$$\text{MOII} = 0.12\%$$

Q. Suppose we have a beaker which is completely filled by water as shown in fig. If we increase the temp by ΔT then analyse the situation if

(a) $\gamma_{\text{liq.}} > \gamma_{\text{beaker}}$ (or $\gamma_{\text{liq.}} > 3\alpha$)



Since $\gamma_{\text{liquid}} > \gamma_{\text{beaker}}$ Hence liquid will expand more than beaker so, some part of liquid will over flow

$$(V_f)_{\text{liquid}} = V_0(1 + \gamma_L\Delta T)$$

$$(V_f)_{\text{beaker}} = V_0(1 + \gamma_b\Delta T) \\ = V_0(1 + 3\alpha\Delta T)$$

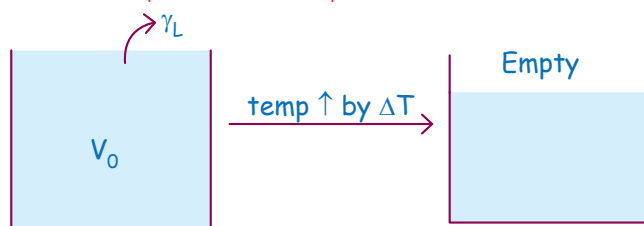
overflow = Kitna volume ka pani bahar gira.

$$= (V_f)_{\text{liquid}} - (V_f)_{\text{beaker}}$$

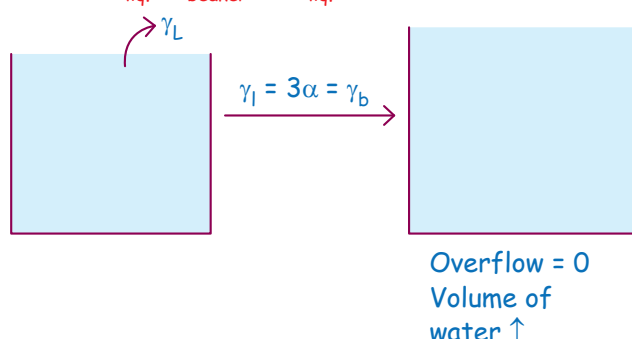
$$= V_0(\gamma_L - 3\alpha)\Delta T = V_0 \gamma_{\text{app}} \Delta T$$

$$\gamma_{\text{app}} = \gamma_L - \gamma_b = \gamma_L - 3\alpha$$

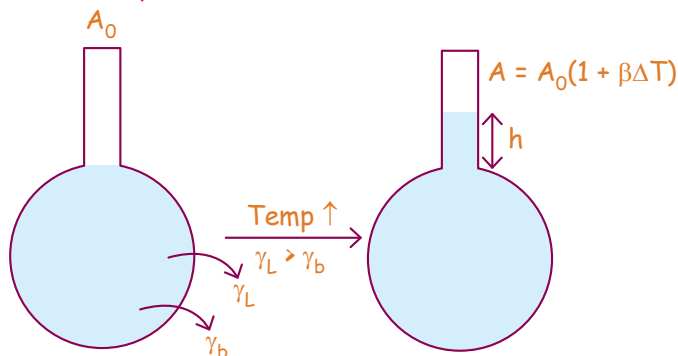
(b) $\gamma_{\text{liq.}} < \gamma_{\text{beaker}}$ (or $\gamma_{\text{liq.}} < 3\alpha$)



(c) $\gamma_{\text{liq.}} < \gamma_{\text{beaker}}$ (or $\gamma_{\text{liq.}} = 3\alpha$)



Q. In following fig. find h after increasing the temp



Sol. $h = \frac{\text{overflow}}{A_f} = \frac{V_0(\gamma_L - \gamma_b)\Delta T}{A_0(1 + \beta\Delta T)}$

$$\beta = 2\alpha$$

$$\gamma_b = 3\alpha$$

Q. Two rods made of different materials are joined end to end and placed between massive walls. The cross-section of the rods are A_1 and A_2 , their lengths L_1 and L_2 , coefficients of linear expansion α_1 and α_2 and the modulus of elasticity of their materials Y_1 and Y_2 respectively. If the rods are heated by $t^\circ\text{C}$. Find the force F which the rods exert on each other.

Sol. Let the first rod expand slightly (say by length δl) and the second rod get compressed by the same amount (since net elongation / compression of the rods is zero).

$\therefore$ Natural increase in length of the first rod (after being heated) when free to expand would have been $\alpha_1 L_1 t$. The expansion allowed is just δl (where $\delta l < \alpha_1 L_1 t$).

$\therefore$ Amount of elongation restricted = $\alpha_1 L_1 t - \delta l$

$$\therefore \text{Strain} = \frac{\text{elongation restricted}}{\text{original length}} = \frac{\alpha_1 L_1 t - \delta l}{L_1 (1 + \alpha_1 t)}$$

Since $\alpha_1 t \ll 1$

$$\therefore 1 + \alpha_1 t \approx 1$$

$$\therefore \text{Strain} = \frac{\alpha_1 L_1 t - \delta l}{L_1}$$

$$\therefore \text{Stress} = \text{strain} \times Y = \left(\frac{\alpha_1 L_1 t - \delta l}{L_1} \right) Y_1$$

$$\text{or } F = \text{stress} \times A = \left(\frac{\alpha_1 L_1 t - \delta l}{L_1} \right) Y_1 A_1$$

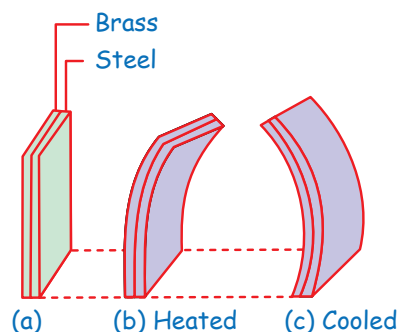
$$\text{Similarly, } F = \left(\frac{+\alpha_2 L_2 t + \delta l}{L_2} \right) Y_2 A_2$$

$$\text{or } \delta l = \alpha_1 L_1 t - \frac{F L_1}{Y_1 A_1} = \frac{F L_2}{Y_2 A_2} - \alpha_2 L_2 t$$

$$\text{or } F = \frac{(\alpha_1 L_1 + \alpha_2 L_2) t}{\left(\frac{L_1}{Y_1 A_1} + \frac{L_2}{Y_2 A_2} \right)}$$

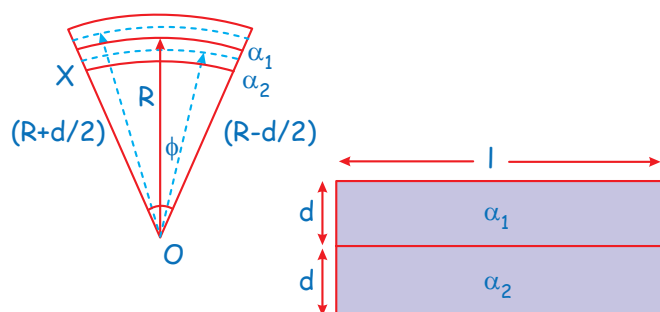
THE BIMETALLIC STRIP

A bimetallic strip is made from two thin strips of metal that have different coefficients of linear expansion as figure shows.



A bimetallic strip (b) when heated and (c) when cooled. Often brass [$\alpha = 19 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$] and steel [$\alpha = 12 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$] are selected. The two pieces are welded or riveted together. When the bimetallic strip is heated, the brass having the larger value of α expands more than the steel. Since, the two metals are bonded together, the bimetallic strips bends into an arc as in part b, with the longer brass piece having a larger radius than the steel piece. When the strip is cooled, the bimetallic strips bends in the opposite direction, as in part c.

Mathematical Solution



$$\left(R + \frac{d}{2}\right) \phi = L_0 (1 + \alpha_1 \Delta\theta)$$

$$\left(R - \frac{d}{2}\right) \phi = L_0 (1 + \alpha_2 \Delta\theta)$$

$$\frac{R + \frac{d}{2}}{R - \frac{d}{2}} = \frac{1 + \alpha_1 \Delta\theta}{1 + \alpha_2 \Delta\theta}$$

From the above equation we can find the value of R.

TEMPERATURE SCALES

For any types of scale

$$\frac{\text{Reading point} - \text{ice point}}{\text{Steam point} - \text{ice point}} = \text{constant}$$

The relation between these scales is given by the following equation.

$$\frac{C - 0^\circ}{100^\circ - 0^\circ} = \frac{K - 273}{373 - 273} = \frac{F - 32}{212 - 32} \Rightarrow \frac{C}{5} = \frac{K - 273}{5} = \frac{F - 32}{9}$$

C = Reading in Celsius

K = Reading in Kelvin

F = Reading in Fahrenheit

CALORIMETRY

Heat

When a hot body is put in contact with a cold one, the former gets colder and the latter warmer. From this observation it is natural to conclude that a certain quantity of heat has passed from the hot body to the cold one. Heat is a form of energy.

$$\begin{array}{ccc} \text{m} & \xrightarrow{\Delta Q = ?} & \text{m} \\ T_1 & & T_2 \\ \Delta Q \propto m & & \\ \alpha \Delta T & & \\ \hline \Delta Q = m S \Delta T \end{array}$$

Here ΔQ = Heat given or taken

m = mass of the body

S = Specific heat constant or specific heat

ΔT = Change in temp

mS = Heat capacity

$$\star \Delta Q = m S \Delta T$$

$$\star dQ = m S dT$$

$$\star \text{In differential form } S = \frac{1}{m} \frac{dQ}{dT}$$

★ S.I. unit of s is joule/kg-kelvin

★ Specific heat of water: $s = 1 \text{ cal/gm}^\circ\text{C}$

★ Specific heat of ice: $s = 0.5 \text{ cal/gm}^\circ\text{C}$

★ Specific heat of steam: $s = 0.46 \text{ cal/gm}^\circ\text{C}$
 $\approx 0.5 \text{ cal/g}^\circ\text{C}$

Q. Find the amount of heat required to raise the temperature of 50 gm water at 20°C to 60°C .
Given $S_{\text{water}} = 1 \text{ cal/gm}^\circ\text{C}$

Sol. 50 gm water $20^\circ\text{C} \xrightarrow{\Delta Q = ?} 50 \text{ gm water } 60^\circ\text{C}$
 $\Delta Q = m S \Delta T = 50 \text{ gm} \cdot \frac{1 \text{ cal}}{\text{gm}^\circ\text{C}} \times (60 - 20)^\circ\text{C} = 200 \text{ cal}$

Q. Find amount of heat release from 50 gm water at 50°C to 50 gm water 20°C .

Sol. $\Delta Q = m S \Delta T$ (magnitude) = $50 \times 1 \times 30 = 1500 \text{ cal}$

Q. 50 gm ice $-20^\circ \xrightarrow{\Delta Q = ?} 50 \text{ gm water } -5^\circ\text{C}$

Sol. $\Delta Q = m S_{\text{ice}} \Delta T = 50 \times \frac{1}{2} \times 15 = 375 \text{ cal}$

Specific Heat Constant (S or C)

It is the amount of energy required to raise the temperature of unit mass of that substance by 1°C (or 1 K) is called specific heat. It is represented by S or C

$$\Delta Q = mS\Delta T \quad d\theta = mSdT$$

$$S = \frac{\Delta Q}{m\Delta T} \quad S = \frac{1}{m} \frac{dQ}{dT}$$

- ★ Thermal capacity = mass $\times$ specific heat = ms
- ★ 1 calorie is defined as amount of heat required to raised the temperature of 1 gm of water from 14.5°C to 15.5°C .
- ★ $1\text{ cal} = 4.182\text{ J} = 4.2\text{ J}$

Molar Heat Capacity

The amount of energy needed to raise the temperature of one mole of a substance by 1°C (or 1 K) is called molar heat capacity.

Molar specific heat capacity, $C = \frac{1}{n} \left(\frac{dQ}{dT} \right)$ where n is no. of moles

- Q.** The specific heat of a substance is given by $C = a + bT$, where $a = 1.12\text{ kJ kg}^{-1}\text{C}^{-1}$ and $b = 0.016\text{ kJ kg}^{-1}\text{C}^{-1}$. The amount of heat required to raise the temperature of 1.2 kg of the material from 280 K to 312 K is:

Sol. Heat required

$$\begin{aligned} Q &= \int_{280}^{312} mC dT = \int_{280}^{312} (1.2)(a + bT) dT \\ &= 1.2 \left[aT + \frac{bT^2}{2} \right]_{280}^{312} \\ &= 1.2 [1.12(312 - 280) + \frac{0.016}{2} \{(312)^2 - (280)^2\}] \\ &= 1.2 [1.12 \times 32 + \frac{0.016}{2} \times (97344 - 78400)] \\ &= 1.2 [35.84 + 0.008 \times 18944] \\ &= 224.8704\text{ KJ} = 225\text{ KJ} \end{aligned}$$

Mechanical Equivalent of Heat

According to Joule, work may be converted into heat and vice-versa. The ratio of work done to heat produced is always constant. $\frac{W}{H} = \text{constant (J)}$
 $\Rightarrow W = JH$

J is called mechanical equivalent of heat.

$J = 4.18\text{ joule/cal}$

WATER EQUIVALENT

- ★ Amount of water that absorbs or gives out the same amount of heat as it done by the body when heated or cooled through 1°C .
- ★ Mass of water having the same thermal capacity as the body is called water equivalent of body.

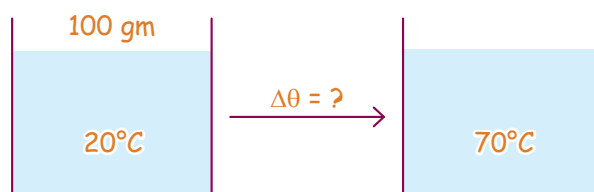
$$\Delta Q = m_b S_b \Delta T = m_w S_w \Delta T \quad (S_w = 1)$$

$$m_w = \frac{m_b S_b}{S_w} = \text{water equivalent} = m_b S_b \text{ (value)}$$

- Q.** Find the amount of heat required to raise the temp of 100 g water in following figure by 50° .

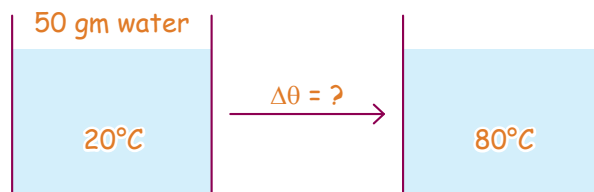
$m_b = 200\text{ gm}$ = mass of beaker

$S_b = 10\text{ cal/gm}^\circ\text{C}$ = specific heat of beaker



Sol. $\Delta\theta = m_w S_w \Delta T + m_b S_b \Delta T$
 $= 100 \times 1 \times 50 + 200 \times 10 \times 50 = 105000\text{ cal}$

- Q.** Find the amount of heat required to raise the temp of 50 gm water by 60°C in following ques. If water equivalent of beaker is 30 gm.



Sol. $\Delta\theta = m_w S_w \Delta T + m_b S_b \Delta T$
 $\Delta\theta = 50 \times 1 \times 60 + 30 \times 60$
 $\Delta\theta = (50 + 30) \times 1 \times 60 = 4800$

Phase Change

- ★ If we supply heat, apart from rising the temperature, it may change the state of body/ substance.

Example: Solid $\longrightarrow$ Liquid
 Liquid $\longrightarrow$ Vapour

★ If m mass of substance changes its state then

$$\Delta\theta \propto m$$

$$\Delta\theta = mL$$

→ Latent heat

Latent Heat or Hidden Heat

When state of a body changes, change of state takes place at constant temperature [melting point or boiling point] and heat released or absorbed is $Q = mL$ where L is latent heat. Heat is absorbed if solid converts into liquid (at melting point) or liquid converts into vapours (at boiling point) and heat is released if liquid converts into solid or vapours converts into liquid.

★ **Latent heat of fusion:** It is the quantity of heat (in kilocalories) required to change its 1 kg mass from solid to liquid state at its melting point.

★ Latent heat of fusion for ice: 80 cal/g.

★ **Latent heat of vaporization:** The quantity of heat required to change its 1 kg mass from liquid to vapour state at its boiling point.

★ Latent heat of vaporisation for water:
= 536 cal/g $\cong$ 540 cal/g (approx)

Chalo bhai, bahut ho gayi definition-definition.
Ab questions ki practice karte hain,
kyunki humare paas limited number
of pages hain.
Isliye ab main questions mein short
language use karunga...
warna publisher wale sir book ki
price badha denge.



Q. Found the amount of heat required to melt 20 g of ice at 0°C .

20 gm ice 0° $\xrightarrow{\Delta\theta = ?}$ 20 gm water 0°C
(solid) (liquid)

Sol. $\Delta Q = mL = 20 \times 80 = 1600 \text{ cal}$

Q. 10 gm water $\xrightarrow{\Delta Q}$ 10 gm vapor
100°C 100°C

Sol. $\Delta Q = mL_v = 10 \times 540 = 5400 \text{ cal}$

Q. 10 gm water $\xrightarrow{\Delta Q = ?}$ 8 gm water 2 gm vapour 100°C

Sol. $\Delta\theta = mL_v = 2 \times 540 = 1080 \text{ cal}$

Q. Find the amount of heat required to change 10 gm water at 20°C to 6 gm water and 4 gm vapour at 100°C .

10 gm water $\xrightarrow{\Delta Q = ?}$ 6 gm water (at 100°C)
20°C 4 gm vapour

Sol.

Sabse pahle water ka temp 20°C se $m\Delta T$ lagakar 100°C water tak le jao, fir $\Delta Q = mL$ lagakar 4 gm water ka phase change kar do.



10 gm water $\xrightarrow{\Delta Q = ?}$ 6 gm water (at 100°C)
 20°C 4 gm vapour
 $\Delta\theta_1 = mS\Delta T$ $\Delta\theta_2 = m'L_v$
 10 gm water $\xrightarrow{\quad}$ 100°C

$$\Delta\theta = mS\Delta T + m'L_v$$

$$\Delta\theta = 10 \times 1 \times 80 + 4 \times 540$$

Q. Find the amount of heat required to change 10 gm ice at -20°C to 6 gm water + 4 gm ice at 0°C .

Sol.

10 gm ice $\xrightarrow{\Delta\theta_{\text{net}}}$ 4 gm ice
 -20°C 6 gm Water at 0°C

$\Delta\theta_1 = mS_{\text{ice}}\Delta T$ 100°C $\Delta\theta_2 = m'L_f$
 $\xrightarrow{\quad}$ 10 gm ice 0°C

$$\Delta\theta_{net} = 10 \times \frac{1}{2} \times 20 + 6 \times 80$$

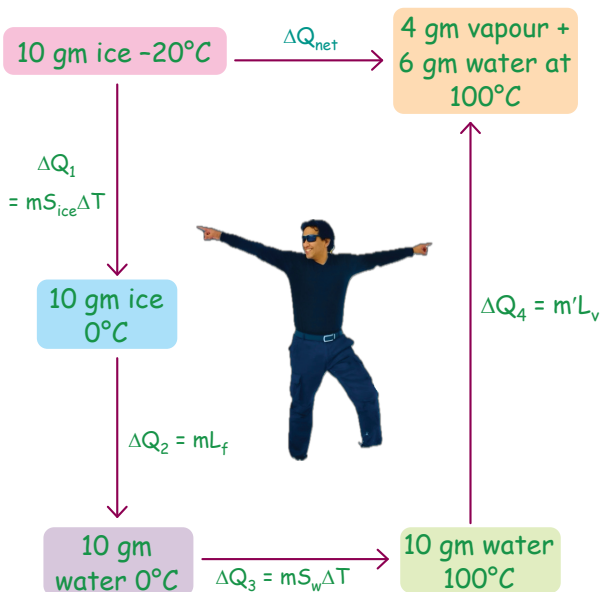
Q. 10 gm ice $\xrightarrow{\Delta Q_{\text{net}} = \Delta Q_1 + \Delta Q_2 + \Delta Q_3}$ 10 gm water 50°
-20°C

Sol. $\Delta Q_{\text{net}} = mS_{\text{ice}}\Delta T_1 + mL_f + mS_w\Delta T_2$

$$= 10 \times \frac{1}{2} \times 20 + 10 \times 80 + 10 \times 1 \times 50$$

Q. Find amount of heat required to change 10 gm ice at -20°C to 4 gm vapour + 6 gm water at 100°C .

Sol.



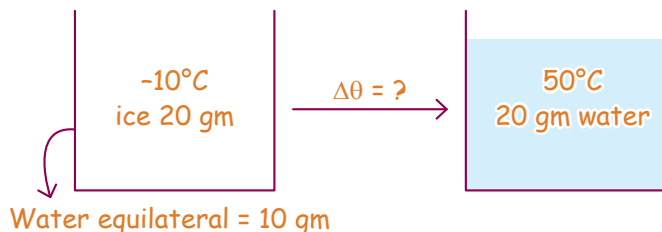
$$\Delta Q_{\text{net}} = 10 \times \frac{1}{2} \times 20 + 10 \times 80 + 10 \times 1 \times 100 + 4 \times 540$$

= Add kar ke khud likh le, ye bhi mai hi karu kya.

Q. 20 gm ice -40°C $\longrightarrow$ 20 gm vapour 100°C

Sol. $\Delta \theta = 20 \times \frac{1}{2} \times 40 + 20 \times 80 + 20 \times 1 \times 100 + 20 \times 540$

Q.



Sol. $\Delta \theta = \left(20 \times \frac{1}{2} \times 10 + 20 \times 80 + 20 \times 1 \times 50 \right) + 10 \times 60$

Q. 40 gm water 50° $\xrightarrow{\Delta Q_{\text{release}} = 40 \times 1 \times 40 = 1600}$ 10 gm water 10°C

Q. 10 gm water 10°C $\longrightarrow$ 10 gm ice 0°C

Sol. Amount of heat release

$$= 10 \times 80 = 800$$

Q. 10 gm Steam 100°C $\xrightarrow{\Delta \theta_{\text{release}} = ?}$ 10 gm water 80°C

$$\Delta Q_1 = 10 \times 540 = 5400 \text{ cal}$$

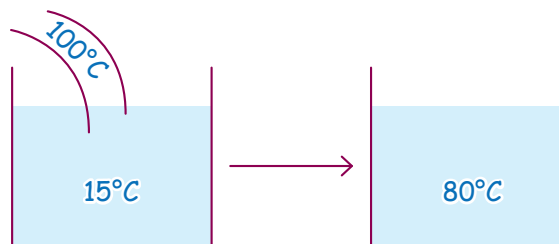
10 gm water 100°C

$$(\Delta Q_2)_{\text{release}} = 10 \times 1 \times 20 = 2000$$

Sol. Amount of heat release = $10 \times 540 + 10 \times 1 \times 20$

Q. Steam at 100°C is passed into 1.1 kg of water contained in a calorimeter of water equivalent .02 kg at 15°C till the temperature of calorimeter and its contents rises to 80°C . What is mass of steam condensed. $L_v = 540 \text{ cal/gm}$.

Sol.



$$m_w S_w (80 - 15) + m_b S_b (80 - 50)$$

$$= m_{\text{steam}} \times 540 + m_{\text{steam}} S_w \times 20$$

$$1.12 \times 10^3 \times 1 \times 65 = m \times 540 + m \times 1 \times 20$$

Q. 500 gm of water at 80°C is mixed with 100 gm steam at 120°C . Find out the final mixture.

Sol. 120°C steam $\longrightarrow$ 100°C steam

$$\text{heat released} = 100 \times 0.46 \times 20 = 920 \text{ cal}$$

$$80^{\circ}\text{C water} \longrightarrow 100^{\circ}\text{C water}$$

$$\text{Req. heat} = 500 \times 1 \times 20 = 10 \text{ kcal}$$

$$100 \text{ gm steam} \longrightarrow 100 \text{ gm water at } 100^{\circ}\text{C}$$

$$\text{heat released} = 100 \times 540 = 54 \text{ kcal}$$

$$\text{Total heat released} = 55 \text{ kcal.}$$

$$\text{Remaining heat} = 55 - 10 = 45 \text{ kcal}$$

$$\text{Now we have } 600 \text{ gm water at } 100^{\circ}\text{C}$$

$$\Rightarrow 45000 = m \times 540 \Rightarrow m = \frac{250}{3} \text{ gm}$$

$$\text{So at last we have } \frac{250}{3} \text{ gm steam and}$$

$$\left(600 - \frac{250}{3} \right) \text{ gm of water}$$

Q. Find amount of heat required if 100 g ice at -10°C is converted into 100 g steam at 120°C .

Sol. -10°C , 100 gm ice $\rightarrow$ 120° 100 gm steam

$$\begin{aligned} Q &= ms\Delta T \\ &= \frac{1}{2} \times 100 \times 10 \\ &= 500 \text{ cal.} \end{aligned}$$

0° , 100 gm ice

$$\begin{aligned} Q &= mL_f \\ &= 100 \times 80 \\ &= 8000 \end{aligned}$$

0° , 100 gm water

$$\begin{aligned} Q &= ms\Delta T \\ &= 100 \times 1 \times 100 \\ &= 10000 \text{ cal.} \end{aligned}$$

100° , 100 gm water

$$\begin{aligned} Q &= mL_v \\ &= 100 \times 540 \\ &= 54000 \text{ cal} \end{aligned}$$

100° , 100 gm Steam

$$\begin{aligned} Q &= ms\Delta T \\ &= 100 \times 0.46 \times 20 \\ &= 920 \text{ cal.} \end{aligned}$$

$$Q_{\text{net}} = 73.42 \text{ kcal.}$$

Q. The specific heat of a substance is given by

$C = a + bT$, where $a = 1.12 \text{ kJ kg}^{-1}\text{C}^{-1}$ and

$b = 0.016 \text{ kJ kg}^{-1}\text{C}^{-1}$. The amount of heat required to raise the temperature of 1.2 kg of the material from 280 K to 312 K is:

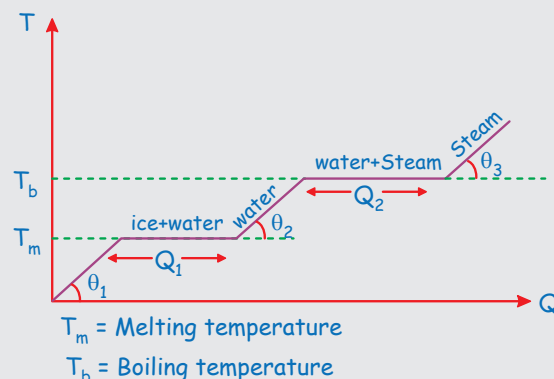
Sol. Heat required

$$\begin{aligned} Q &= \int_{280}^{312} mC dT = \int_{280}^{312} (1.2)(a + bT) dT \\ &= 1.2 \left[aT + \frac{bT^2}{2} \right]_{280}^{312} \\ &= 1.2 \left[1.12 (312 - 280) + \frac{0.016}{2} \{ (312)^2 - (280)^2 \} \right] \\ &= 1.2 \left[1.12 \times 32 + \frac{0.016}{2} \times (97344 - 78400) \right] \\ &= 1.2 [35.84 + 0.008 \times 18944] \\ &= 224.8704 \text{ KJ} \\ &= 225 \text{ KJ} \end{aligned}$$

Heating Curve

Heating Curve

If to a given mass (m) of a solid, heat is supplied at constant rate and a graph is plotted between temperature and time as shown in figure is called heating curve.



If heat is supplied at constant rate

$$dQ = msdT$$

$$\text{Slope} = \frac{dT}{dQ} = \frac{1}{ms}$$

$$\text{Since } s_{\text{ice}} = s_{\text{steam}} = 0.5 \text{ cal/g}^{\circ}\text{C}$$

$$\text{and } s_{\text{water}} = 1 \text{ cal/g}^{\circ}\text{C}$$

$$\Rightarrow \theta_1 = \theta_3 > \theta_2 \text{ and } Q_2 > Q_1$$

Q. Suppose we have m mass of ice at -40°C and we are supplying heat at constant rate P . Draw the temperature vs time graph

Sol. When temperature of ice is increasing.

$$d\theta = ms dT$$

$$\frac{d\theta}{dt} = ms \frac{dT}{dt} = P$$

$$\text{Slope} = \frac{dT}{dt} = \frac{P}{ms} = \frac{P}{ms_{\text{ice}}} = \text{Constant}$$

When temperature of water is increasing

$$d\theta = ms dT$$

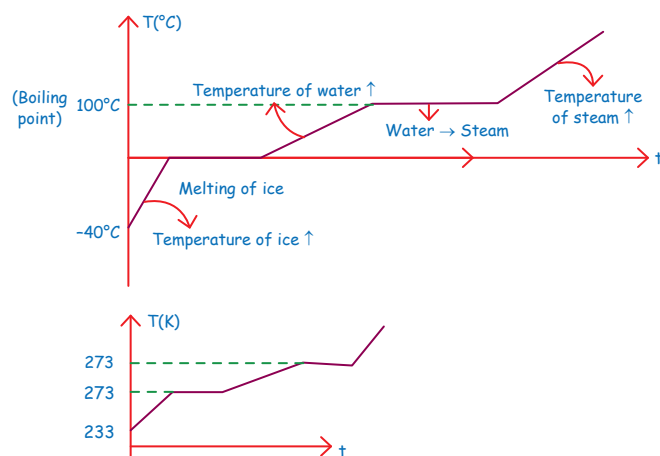
$$\frac{d\theta}{dt} = ms \frac{dT}{dt} = P = \text{Constant}$$

$$\text{Slope} = \frac{dT}{dt} = \frac{P}{ms} = \frac{P}{ms_{\text{water}}} = \text{Constant}$$

$$\therefore s_{\text{water}} > s_{\text{ice}}$$

$$(\text{Slope})_{\text{water}} < (\text{Slope})_{\text{ice}}$$

$$\frac{\tan \theta_2}{\tan \theta_1} = \frac{(\text{Slope})_{\text{water}}}{(\text{Slope})_{\text{ice}}} = \frac{S_i}{S_w} = \frac{1/2}{1} = \frac{1}{2}$$



Principle of Calorimetry

When two bodies at different temperature are allow to share heat, they attain a common temperature, and if assume that no heat is received from or given to any body outside the system and there is no chemical action involved in the process of sharing then heat given by one body

= heat loss by another body

Q. 40 gm water + 60 gm water $\longrightarrow$ T_f ?
 20°C 40°C of mixture
 (A) (B)

Sol. $\Delta Q_1 + \Delta Q_2 = 0$ (from calorimetry principle)

$$40 \times 1 \times (T_f - 20) + 60 \times 1 \times (T_f - 40) = 0$$

$$4T_f - 80 + 6T_f - 240 = 0$$

$$T_f = 32^\circ\text{C}$$

Q. 40 gm + 60 gm + 40 gm $\longrightarrow$ T_f ?
 water water water of mixture.
 20°C 40°C 50°C

Sol. $\Delta\theta_1 + \Delta\theta_2 + \Delta\theta_3 = 0$ (from calorimetry principle)

$$40 \times 1 \times (T_f - 20) + 60 \times 1 \times (T_f - 40) + 40 \times 1 \times (T_f - 50) = 0$$

$$T_f = 37.1^\circ\text{C}$$

Q. A calorimeter of heat capacity 100 J/K is at room temperature of 30°C 100 gm of water at 40°C specific heat 4200 J/kg/K is poured into the calorimeter. What will be temperature of water in calorimeter.

Sol. $\Delta\theta_1 + \Delta\theta_2 = 0$

$$m_b S_b (T_f - 30) + m_w S_w (T_f - 40) = 0$$

$$100 (T_f - 30) + \frac{100}{1000} \times 4200 \times (T_f - 40) = 0$$

$$T_f = 38^\circ\text{C}$$

Q. 10 gm + 30 gm $\longrightarrow$ T_f ?
 $S_A = 20 \text{ cal/gm}^\circ\text{C}$ $S_B = 25 \text{ cal/gm}^\circ\text{C}$
 10°C 40°C

(A)

(B)

Sol. $\Delta\theta_1 + \Delta\theta_2 = 0$

$$10 \times 20 \times (T_f - 10) + 30 \times 25 \times (T_f - 40) = 0$$

$$T_f - 30) + \frac{32000}{950} \approx 33.68^\circ\text{C}$$

Q. In a container of negligible heat capacity, 200 gm ice at 0°C and 100 gm steam at 100°C are added to 200 gm of water that has temperature 55°C . Assume no heat is lost to the surroundings and the pressure in the container is constant 1.0 atm. (Latent heat of fusion of ice = 80 cal/gm, Latent heat of vaporization of water = 540 cal/gm, Specific heat capacity of ice = 0.5 cal/gm-K, Specific heat capacity of water = 1 cal/gm-K)

(a) What is the final temperature of the system?

(b) At the final temperature, mass of the total water present in the system, is

(c) Amount of the steam left in the system, is equal to

Sol. As steam has comparatively large amount of heat to provide in the form of latent heat we check what amount of heat is required by the water and ice to go up to 100°C , that is

$$(m_i L + m_i S_w \Delta T) + m_w \cdot S_w \cdot \Delta T$$

$$= [(200 \times 80) + (200 \times 1 \times 100)] + (200 \times 1 \times 45)$$

$$= 45,000 \text{ cal.}$$

That is given by m mass of steam, then

$$m_s L = 45,000$$

$$m_s = \frac{45,000}{540} = \frac{500}{6} = 83.3 \text{ gm}$$

Therefore 83.3 gm steam converts into water of 100°C .

$$\text{Total water} = 200 + 200 + 83.3 = 483.3 \text{ gm}$$

$$\text{Steam left} = 16.7 \text{ gm.}$$

HEAT TRANSFER

Mode of Heat Transfer

Heat is a form of energy which transfer from a body at higher temperature to a body at lower temperature. The transfer of heat from one body to another may take place by any one of the following modes:

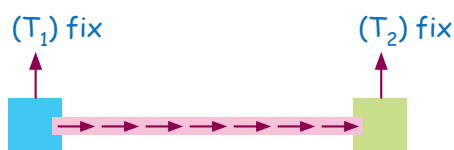
Conduction	Convection	Radiation
Heat Transfer due to Temperature difference	Heat transfer due to density difference	Heat transfer with out any medium
Due to free electron or vibration motion of molecules	Actual motion of particles	Electro-magnetic radiation
Heat transfer in solids body (in mercury also)	Heat transfer in fluids (Liquid + gas)	All
Slow process	Slow process	Fast process (3×10^8 m/sec)
Irregular path	Irregular path	Straight line (like light)

Conduction

The process by which heat is transferred from hot part to cold part of a body through the transfer of energy from one particle to another particle of the body without the actual movement of the particles from their equilibrium positions is called conduction.

Steady state: When temperature of the each cross-section of the bar becomes constant through different for different cross-sections is called steady state.

Suppose we have a rod whose temperature of one end is fixed at T_1 and another end is fix at T_2 . If $T_1 > T_2$ heat will start flowing from T_1 to T_2 .



Heat Transfer

After steady state we have observed rate of flow of heat $\frac{dQ}{dt}$

$$\frac{dQ}{dt} \propto \Delta T \quad (\text{Temp. differece})$$

$$\propto A \quad (\text{Cross - sectional Area})$$

$$\propto \frac{1}{l} \quad (l : \text{length of material})$$

$$\frac{dQ}{dt} = \frac{KA\Delta T}{l}$$

K = Coefficient of thermal conductivity

$$\frac{dQ}{dt} = \frac{KA\Delta T}{l}$$

(SI unit : W/mk
ie watt/meter kelvin)

★ In differential form this heat equation can be written as

$$\frac{dQ}{dt} = -KA \frac{dT}{dx}$$

Where $\frac{dT}{dx}$ is called temperature gradient

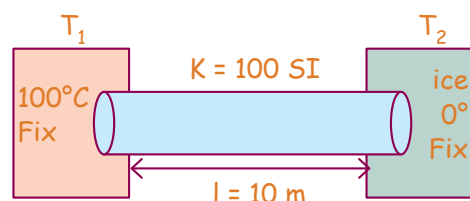
Conduction ka ye eklauta formula hai jis par pura chgapter tika hai. $\frac{dQ}{dt}$ means rate of flow of heat, bole toh per second kitani heat transfer ho rahi hai.

$$\star \frac{dQ}{dt} = \frac{KA\Delta T}{l} \quad (\text{Magnitude})$$

$$\star \frac{dQ}{dt} = -KA \frac{dT}{dx}$$



Q. Find rate of flow heat and rate of melting of ice in following case.



Sol. Rate of flow of heat = $\frac{dQ}{dt} = \frac{KA\Delta T}{l}$

$$\frac{dQ}{dt} = \frac{100 \times 2 \times 100}{10} = 2000 \text{ J/sec}$$

$$\frac{dQ}{dt} = 2000 = \frac{d}{dt} (ML)$$

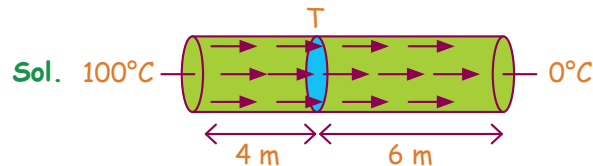
$$\Delta\theta = mL$$

$$\Rightarrow 2000 = L \frac{dm}{dt}$$

Now you can find $\frac{dm}{dt}$ which is rate of melting of ice but $L = 80$ put mat kar dena bcz yahan $\frac{dQ}{dt}$ joule per sec mein hai so, L ki value hi joule/kg mein put karna warna pitai hogi.



Q. Find the temperature at a distance 4 m from hot end in above case.



At steady state $\frac{dQ}{dt}$ is same for all cross-section area.

$$\begin{aligned}\frac{dQ}{dt} &= \frac{KA(100 - 0)}{10} \\ &= \frac{KA(100 - T)}{4} = \frac{KA(T - 0)}{6}\end{aligned}$$

Now solve and get $\Rightarrow T = 60^\circ\text{C}$

SKC

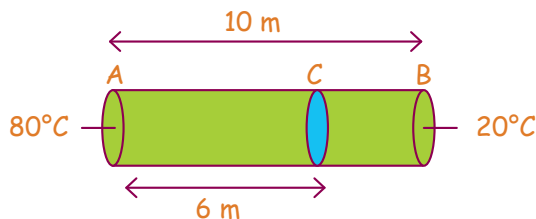
10 m me drop aaya $\longrightarrow 100$ ka

1 m me drop aayega $\longrightarrow \frac{100}{10} = 10$ ka

So, 4 m me drop aayega $\longrightarrow 10 \times 4 = 40$ ka

Hence, 40°C ka drop aaya so, $T = 100 - 40 = 60^\circ\text{C}$

Q. Find temperature at C.



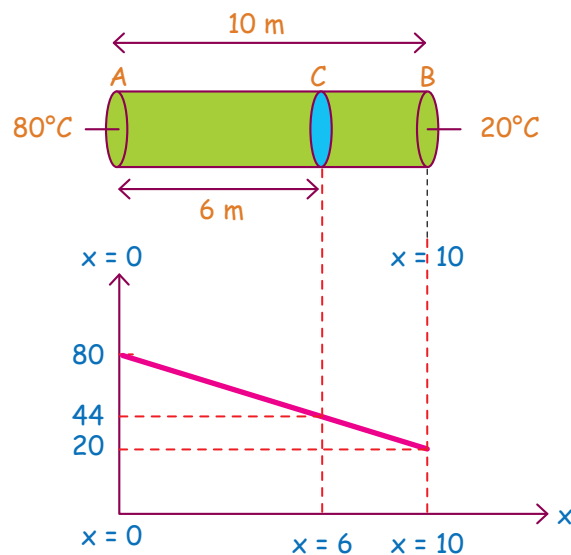
Sol. 10 m $\longrightarrow 60^\circ\text{C}$ (fall)
4 m $\longrightarrow \frac{60}{10} \times 6 = 36^\circ\text{C}$
 $T_B = 80 - 36 = 44$

$$\frac{dQ}{dt} = \frac{KA(80 - 20)}{10} = \frac{KA(80 - T)}{6} = \frac{KA(T - 20)}{4}$$

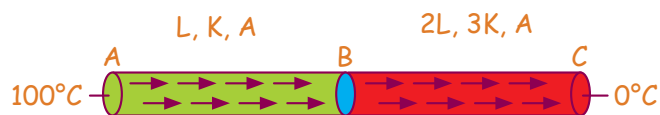
Solve any two eqn & get the answer

$T = 44$

Draw the graph between temperature and x if left end of rod at 80°C is at $x = 0$.



Q. Two different rod are connected to each other as shown in fig. Find temperature at junction B after steady state.



Sol.
$$\frac{dQ}{dt} = \frac{KA(100 - T)}{L} = \frac{3KA(T - 0)}{2L}$$

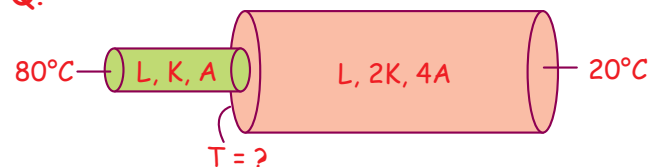
Same

$$200 - 2T = 3T$$

$$T = 40^\circ\text{C}$$

There is no accumulation of heat anywhere

Q.



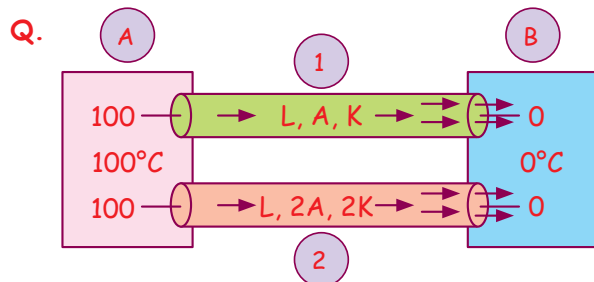
There is no accumulation of heat anywhere

$\left(\frac{dQ}{dt} \text{ har jagah} \rightarrow \text{same}\right)$

Sol.
$$\frac{dQ}{dt} = \frac{KA(80 - T)}{L} = \frac{2K \cdot 4A(T - 20)}{L}$$

$$80 - T = 8T - 160$$

$$T = \frac{240}{9} = \frac{80}{3}$$



Sol.

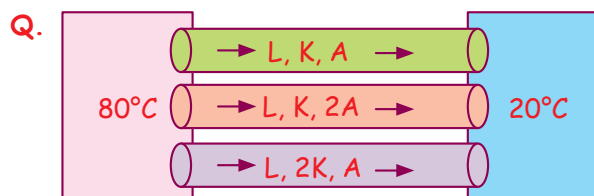
$$\left(\frac{dQ}{dt}\right)_{\text{upar}} = \frac{KA(100 - 0)}{L}$$

$$\left(\frac{dQ}{dt}\right)_{\text{neeche}} = \frac{(2K)(2A)(100 - 0)}{L}$$

$$\left(\frac{dQ}{dt}\right)_{\text{net } A \rightarrow B} = (\text{Upar}) + (\text{neeche})$$

$$= \frac{KA(100 - 0)}{L} + \frac{2K \cdot 2A(100 - 0)}{L}$$

$$= \frac{5KA}{L} \times 100$$



Sol.

$$\left(\frac{dQ}{dt}\right)_{\text{net}} = \frac{KA(80 - 20)}{L} + \frac{K(2A)(80 - 20)}{L} + \frac{2KA(80 - 20)}{L}$$

Q. Suppose we have a rod whose temperature of one end is fixed at T_1 at $x = 0$ and another end is fixed at T_2 at $x = L$. If $T_1 > T_2$ then draw the graph between T vs x and analysis the situation.

Sol. $\therefore \frac{dQ}{dt} = -KA \frac{dT}{dx}$; $(T_1) \text{ fix}$ $(T_2) \text{ fix}$

So, $\frac{dT}{dx} = -\frac{(dQ/dt)}{KA}$

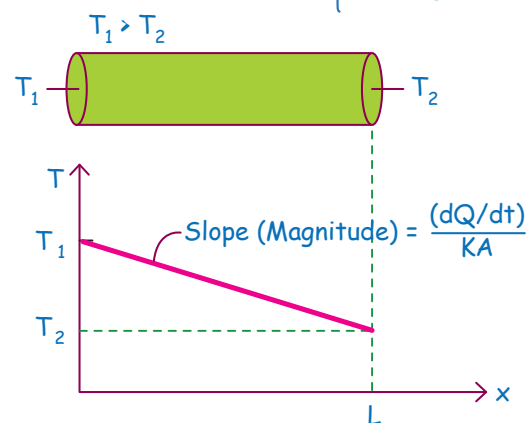
$\frac{dQ}{dt}$ is same for all cross-section area and if we plot graph between T vs x then $\frac{dT}{dx} = \text{slope}$

So, $\frac{dT}{dx} = -\frac{(dQ/dt)}{KA} = \text{slope}$

Slope $\propto \frac{1}{KA}$

If K and A is same everywhere $\Rightarrow$ slope will be constant graph will be straight line and after

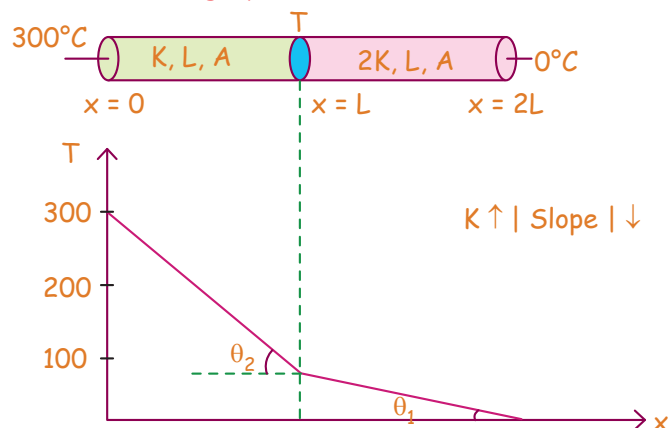
solving you can get $T(x) = \left\{ T_1 - \frac{(T_1 - T_2)}{L}(x) \right\}$



$$\frac{dQ}{dt} = -KA \frac{dT}{dx}$$

$$\frac{dT}{dx} = -\frac{(dQ/dt)}{KA} = \text{Slope (Constant)}$$

Q. Draw the graph between T vs x .



Sol. For temperature at junction solve

$$\frac{dQ}{dt} = \frac{KA}{L} (300 - T) = \frac{2KA}{L} (T - 0)$$

$$T = 100$$

In sawalon ko ya to aise, ya mushkil sawalon ko karne ka ek mast tareeka hai. Batatu chalo rehne dete hain.



Ab tak humne dekha



$$\frac{dQ}{dt} = \frac{KA\Delta T}{l} = \frac{\Delta T}{l/KA}$$

$$\left(\frac{dQ}{dt} = \text{heat current}\right) = i \text{ (let)}$$

Ab in sawalon ko karne ka ek naya aur mast concept padhenge... isme thoda sa 10th class ka current electricity use hoga, samajhne ki koshish karna.

$$\frac{dQ}{dt} = i = \frac{\Delta T}{l/KA} = \frac{\Delta T}{R_{th}} \Rightarrow i = \frac{\Delta V}{R} \text{ (Yaad karo 10th me padha tha.)}$$

Jaise CE me V (potential) hota tha same aese hi yahan temperature hai, jaise wahan ΔV hota tha yahan ΔT hai wahan high potential se low potential electric current flow karta tha yahan high temperature to low temperature heat (heat current) flow karti hai means

Heat flow through a conducting rod	Current flow through a resistance
Heat current $H = \frac{dQ}{dt}$ = rate of heat flow	Electric current $i = \frac{dq}{dt}$ = rate of charge flow
$H = \frac{\Delta T}{R}$	$i = \frac{\Delta V}{R}$

$$V \longrightarrow T$$

$$\Delta V \longrightarrow \Delta T$$

$$i \longrightarrow \frac{dQ}{dt}$$

$$i = \frac{\Delta V}{R} \longrightarrow \frac{dQ}{dt} = \frac{\Delta T}{l/KA} = \frac{\Delta T}{R_{th}}$$

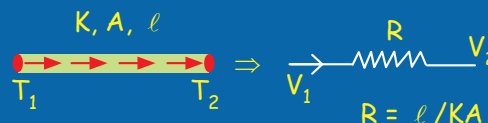
$$R = \rho \frac{l}{A} = \frac{l}{\sigma A} \longrightarrow R_{th} = \frac{l}{KA}$$

σ = electrical conductivity

K = thermal conductivity

SKC

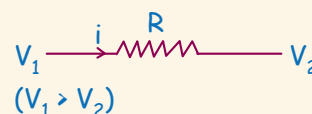
Arey! Samajh rahe ho na... Mudde ki baat ye hai ki conducting rod ko hum aese resistance se replace kar sakte hai jiske resistance ki value l/KA hogi



SKC bole to rod hatao, wire banao.



10th class revision

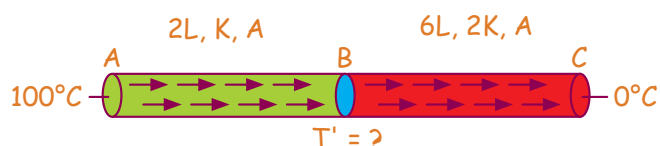


$$i = \frac{V_1 - V_2}{R} = \frac{\Delta V}{R} = i$$

$$\star 100^\circ\text{C} \xrightarrow{I, K, A} 0^\circ\text{C} \equiv \xrightarrow{i} \xrightarrow{R = l/KA} 100\text{ V} \xrightarrow{} 0\text{ V}$$

$$i = \frac{100 - 0}{R} = \frac{100}{l/KA}$$

Q. Find the heat current and temp T' at junction



Sol. Method-I:

$$\frac{dQ}{dt} = \frac{KA(100 - T')}{2L} = \frac{2KA(T' - 0)}{6L}$$

Solve and get

Method-II: (SKC)

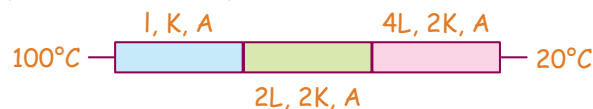
Rod hatao, wire banao Let $R = l/KA$



$$i = \frac{100}{5R} = \frac{20}{R} = \frac{20}{\frac{l}{KA}}$$

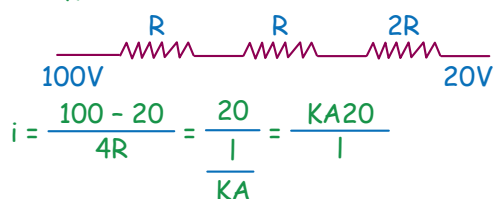
$$T' = 60^\circ\text{C}$$

Q. Find heat current.

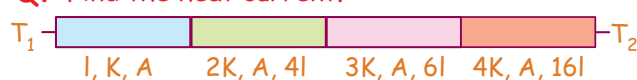


$$\frac{d\theta}{dt} = i = ?$$

Sol. $R = \frac{l}{KA}$ (let) so, above problem is analogous to



Q. Find the heat current.



Sol. $R = \frac{l}{KA}$ (let) so, above problem is analogous to

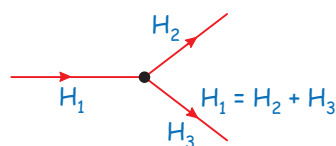
$$\frac{dQ}{dt} = i = \frac{T_1 - T_2}{R_{eq}}$$

$$= \frac{T_1 - T_2}{R + 2R + 2R + 4R} = \frac{T_1 - T_2}{9 \left(\frac{l}{KA} \right)}$$

• where $R = \frac{l}{KA}$

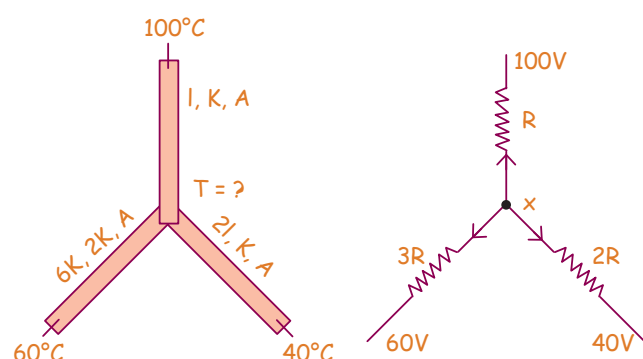
JUNCTION LAW

According to the Junction law the sum of all the heat current directed towards a point is equal to the sum of all the heat currents directed away from the points.



Similarly in current electricity we will study at the junction $i_1 = i_2 + i_3$

Q. Find temperature at the junction

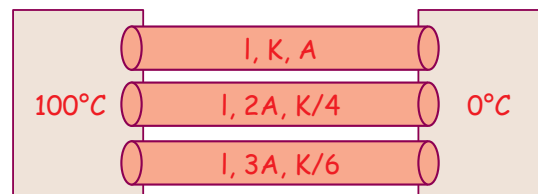


Heat Transfer

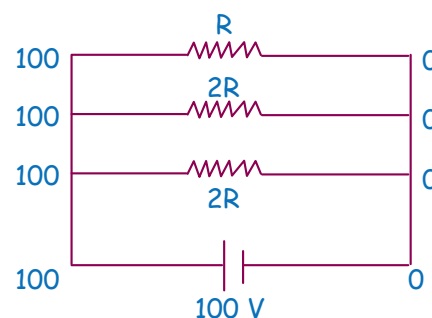
Sol. $i_1 + i_2 + i_3 = 0$ (Kirchoff's Junction Rule)

$$\frac{x - 100}{R} + \frac{x - 60}{3R} + \frac{x - 40}{2R} = 0 \text{ form this you can evaluate } x \text{ (ie. temp. at Junction)}$$

Q. Find net rate of heat flow.



Sol.



$$R_1 = \frac{l}{KA} = R \text{ (let)}$$

$$R_2 = \frac{l}{\frac{K}{4} 2A} = \frac{2l}{KA} = 2R$$

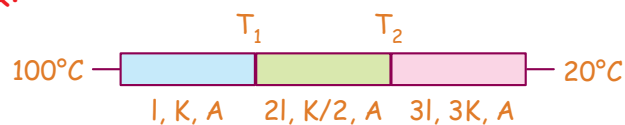
$$R_3 = \frac{l}{\frac{K}{6} 3A} = 2R$$

$$\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{2R} + \frac{1}{2R}$$

$$R_{eq} = \frac{R}{2}$$

$$i = \frac{\Delta V}{R_{eq}} = \frac{100 - 0}{\frac{R}{2}}$$

Q.



Find T_1 , T_2 and $\frac{d\theta}{dt}$.



Sol.

$$i = \frac{80}{6R} = \frac{80KA}{6l}$$

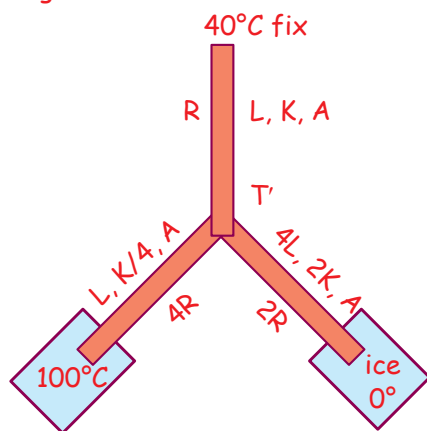
$$i = \frac{80}{6R} = \frac{100 - V_1}{R}$$

$$V_1 = 100 - \frac{80}{6} = \frac{520}{6}$$

$$i = \frac{80}{6R} = \frac{100 - V_2}{5R}$$

$$V_2 = 100 - \frac{400}{6} \Rightarrow V_2 = \frac{200}{6}$$

Q. Find temperature at junction and rate of melting of ice.



Sol.
$$\frac{T' - 40}{R} + \frac{T' - 100}{4R} + \frac{T' - 0}{2R} = 0$$

$$4(T' - 40) + (T' - 100) + 2T' = 0$$

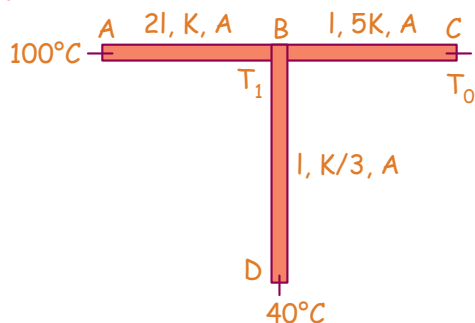
$$7T' = 260$$

$$T' = \frac{260}{7}^{\circ}\text{C}$$

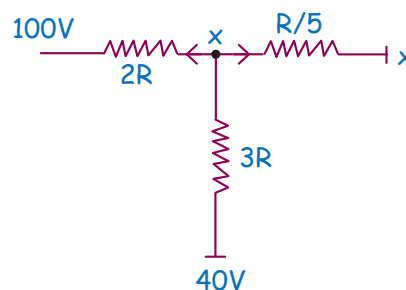
Rate of melting of ice =

$$\left(\frac{d\theta}{dt}\right)_{\text{ice}} = \frac{2KA(T' - 0)}{4L} = L_f \left(\frac{dm}{dt}\right)$$

Q. If heat current in rod BC is zero find $T_c = :$



Sol. Voltage - current equivalent diagram.

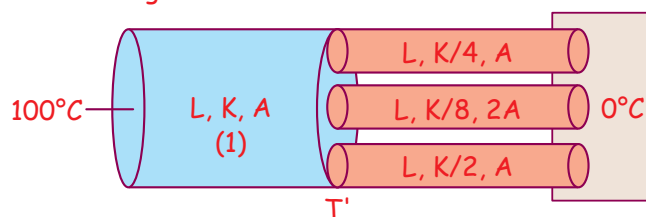


$$\frac{x - 100}{2R} + \frac{x - 40}{3R} + 0 = 0$$

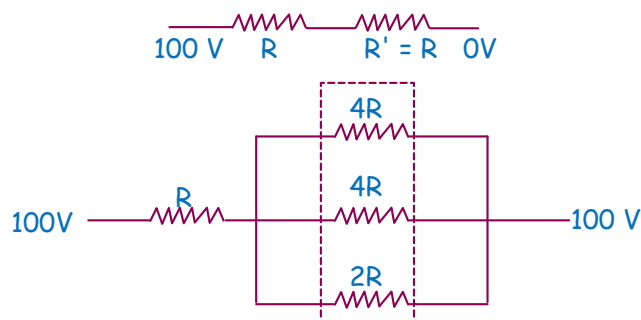
$$3x - 300 + 2x - 80 = 0$$

$$x = 76 \quad 76^{\circ}\text{C} \quad x = T_B = T_C = 76^{\circ}\text{C}.$$

Q. Find T' and net rate of flow of heat $\left(\frac{dQ}{dt}\right)$ through rod '1'.



Sol. $R = \frac{L}{KA}$ (let) then the current - voltage equivalent:



$$\frac{dQ}{dt} = i = \frac{100 - 0}{2R} = \frac{50}{R} = \left(\frac{L}{KA}\right)$$

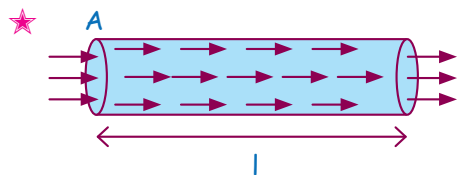
$$T' = 50$$

Ab hum alag-alag body ka thermal resistance nikalna seekhenge...jo aapko 12th class ke current electricity mein as it is dekhne ko milega.

I will suggest, agar possible ho, to isse yaad kar lena, waise zyada mushkil nahi hai.



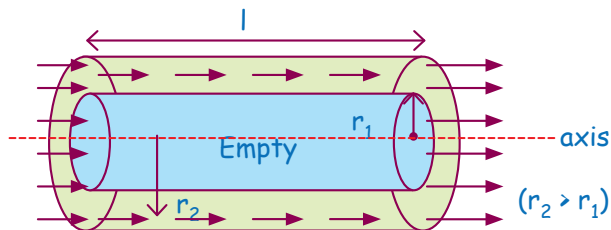
Thermal Resistance



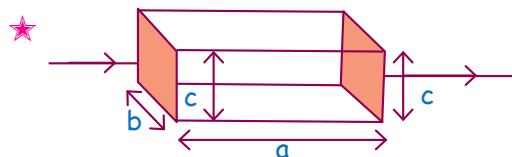
Solid cylinder (radius 'r')

$$R = \frac{l}{KA} = \frac{l}{K\pi r^2}$$

- ★ Hollow cylinder of inner radius r_1 and outer radius r_2 and heat is flowing along the axis.

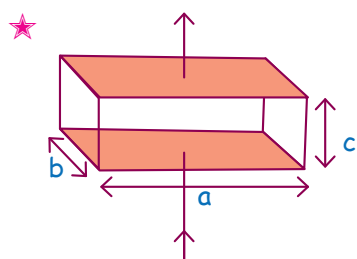


$$R = \frac{l}{KA} = \frac{l}{K\pi(r_2^2 - r_1^2)}$$



$$R_{eq} = \frac{l}{K(A)} = \frac{a}{Kbc}$$

cross section.



direction of heat flow.

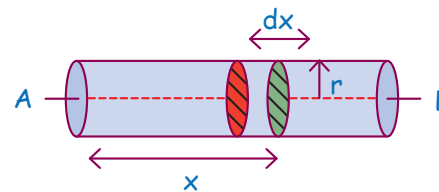
$$R_{eq} = \frac{c}{Kab}$$

- Q. Thermal conductivity varies with length (x) then find, R_{AB}

$$K = 2x + 3$$



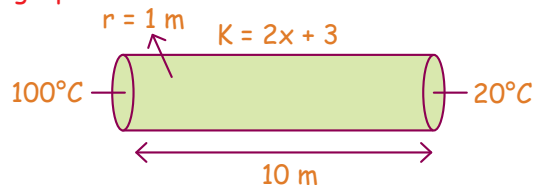
Sol.



$$R_{AB} = \int dR = \int_0^l \frac{dx}{(2x+3)\pi r^2} \quad \left(dR = \frac{dx}{KA}\right)$$

$$R_{AB} = \frac{1}{2\pi r^2} \ln(2x+3) \Big|_0^l = \frac{1}{2\pi r^2} \ln\left(\frac{2l+3}{3}\right)$$

- Q. Find rate of heat flow through the solid cylinder as shown in fig and R_{eq} if thermal conductivity of material varies as $K = 2x + 3$ length of the rod is 10 m and radius 1 m. Also draw the T vs x graph.

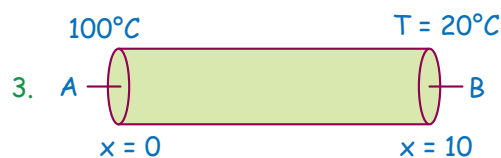


- Sol. 1. $R_{eq} = ?$

$$\int dR = \int_0^{10} \frac{dx}{(2x+3)\pi r^2}$$

$$R_{eq} = \frac{1}{2\pi r^2} \ln\left(\frac{23}{3}\right)$$

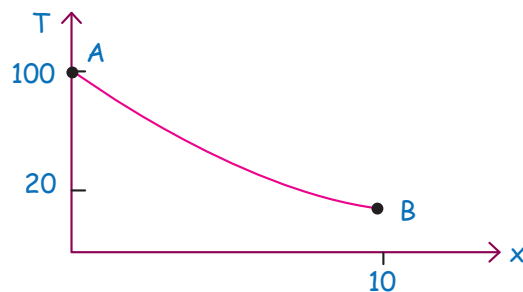
$$2. \left(\frac{d\theta}{dt}\right) = i = \frac{100 - 20}{R_{eq}}$$



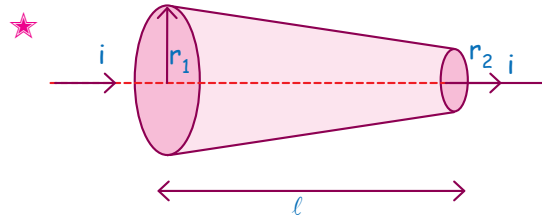
$$i = -KA \frac{dT}{dx}$$

$$\frac{dT}{dx} = \frac{-i}{KA} = \frac{-i}{(2x+3)A}$$

$x \uparrow$ | Slope | $\downarrow$



Very Important

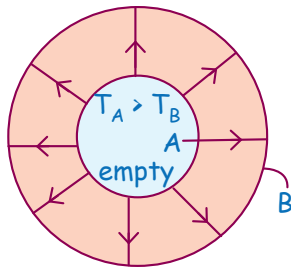


$$R_{eq} = \frac{\ell}{K\pi r_1 r_2}$$



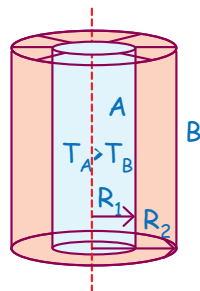
$$R = \frac{\ell}{K\pi r^2}$$

★ Hollow sphere (Heat flow Radially out)



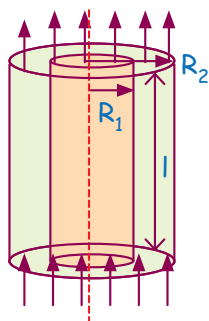
$$R_{eq} = \frac{R_2 - R_1}{K4\pi R_1 R_2}$$

★ Hollow cylinder (Heat flow radially out word)



$$R_{AB} = \frac{1}{K2\pi\ell} \ln \left(\frac{R_2}{R_1} \right)$$

★ Direction of heat flow



$$R = \frac{\ell}{K\pi(R_2^2 - R_1^2)}$$

$$R = \frac{\ell}{KA} \equiv R = \frac{1}{\sigma} \frac{\ell}{A} \quad \text{C.E.}$$

$$R = \rho \frac{\ell}{A}$$

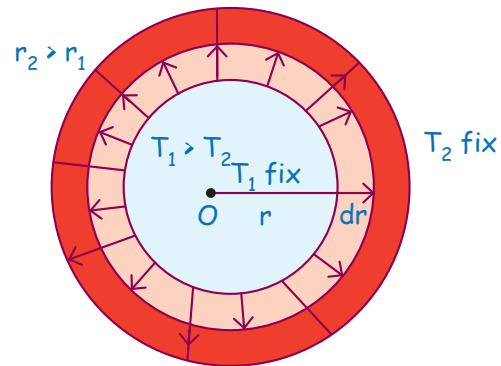
$$\frac{\ell}{K} \equiv \rho = \frac{1}{\sigma}$$

Hollow Sphere (thermal resistance derivation)

$$\int dR = \int_{R_1}^{R_2} \frac{dr}{K(4\pi r^2)} \quad \text{element will be a sphere of thickness 'dr' and radius 'r'}$$

$$R_{eq} = \frac{1}{4\pi K} \int_{R_1}^{R_2} \frac{1}{r^2} dr$$

$$R_{eq} = \frac{1}{4\pi K} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = \frac{1}{4\pi K} \left(\frac{R_2 - R_1}{R_1 R_2} \right)$$

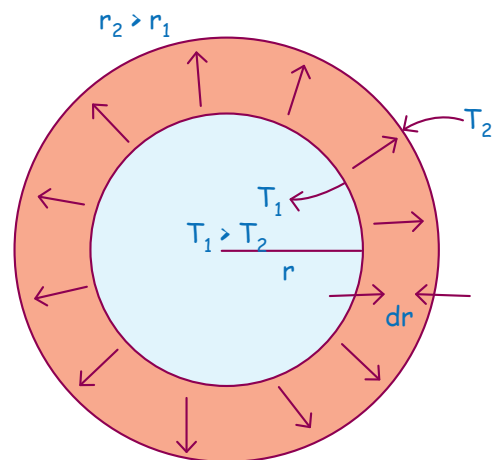


$$i = \frac{d\theta}{dt} = \frac{T_1 - T_2}{R_{eq}}$$

$$\int dR = \int_{r_1}^{r_2} \frac{dr}{K4\pi r^2}$$

$$R_{eq} = \frac{1}{4\pi r} - \frac{1}{r} \Big|_{r_1}^{r_2}$$

$$\Rightarrow R_{eq} = \frac{1}{4\pi K} \left(\frac{r_2 - r_1}{r_1 r_2} \right)$$



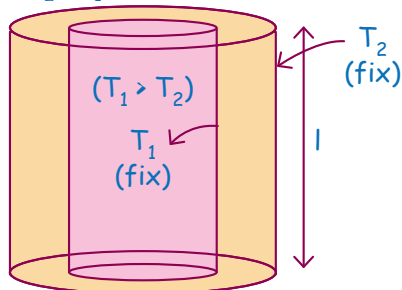
$$i = \frac{T_1 - T_2}{R_{eq}} = \frac{(T_1 - T_2)}{(r_2 - r_1)} 4\pi R_1 R_2$$

Hollow Cylinder (thermal resistance derivation)

$$\int dR = \int_{r_1}^{r_2} \left(\frac{dr}{K(2\pi r l)} \right) = \frac{1}{2\pi k l} \int_{r_1}^{r_2} \frac{dr}{r}$$

$$R_{eq} = \frac{1}{2\pi k l} \ln \left(\frac{r_2}{r_1} \right)$$

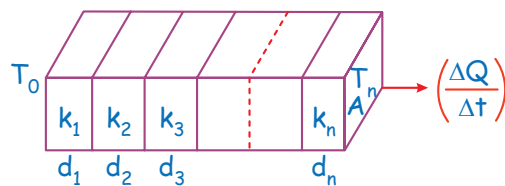
$$(r_2 > r_1)$$



$$i = \frac{T_1 - T_2}{R_{eq}}$$

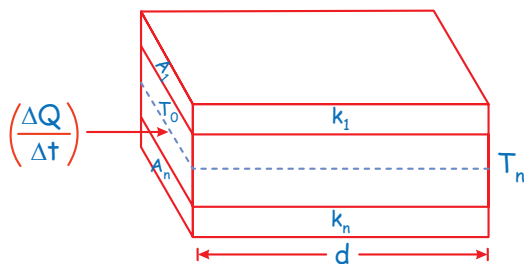
IMPORTANT RESULT

Layers in Series



$$\frac{\Delta Q}{\Delta t} = \frac{A(T_0 - T_n)}{\frac{d_1}{k_1} + \frac{d_2}{k_2} + \dots + \frac{d_n}{k_n}}$$

Layers in Parallel



$$\frac{\Delta Q}{\Delta t} = \frac{(T_0 - T_n)(k_1 A_1 + k_2 A_2 + \dots + k_n A_n)}{d}$$

Effective Thermal Conductivity in Series Combination

$$k_{eq} = \frac{d}{\frac{d_1}{k_1} + \frac{d_2}{k_2} + \frac{d_3}{k_3} + \dots + \frac{d_n}{k_n}}$$

Effective Thermal Conductivity in Parallel Combination: [Length will be Uniform]

$$k_{eq} = \frac{\frac{1}{A}}{\frac{1}{k_1 A_1} + \frac{1}{k_2 A_2} + \dots + \frac{1}{k_n A_n}}$$

Q. Two identical metal of thermal conductivities K_1 and K_2 respectively are connected in series. The effective thermal conductivity of the combination is:

Sol. Formula of thermal resistance, $R = \frac{l}{KA}$

$$R_{eq} = R_1 + R_2 = \frac{l}{K_1 A} + \frac{l}{K_2 A}$$

$$= \frac{l}{A} \left(\frac{1}{K_1} + \frac{1}{K_2} \right) = \frac{2l}{KA}$$

$$\Rightarrow K = \frac{2K_1 K_2}{K_1 + K_2}$$

Q. Find 1. R_{eq} 2. K_{eq}

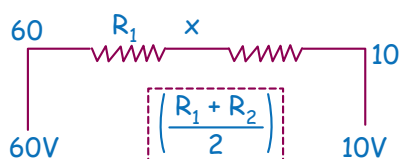
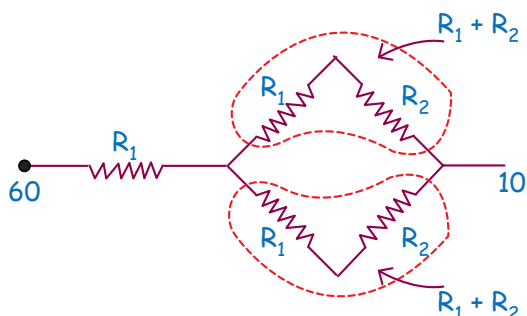
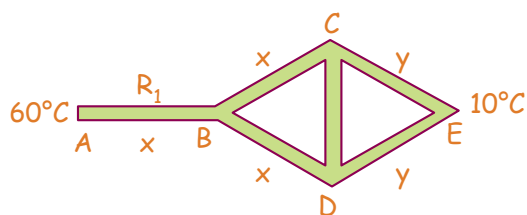
Sol. 1. $R_{eq} = \frac{l}{K_1 A} + \frac{l}{K_2 A}$ $R_{eq} = \frac{2l}{K_{eq} A}$

$$\frac{l}{K_1 A} + \frac{l}{K_2 A} = \frac{2l}{K_{eq} A} \quad \frac{K_1 + K_2}{K_1 K_2} = \frac{2}{K_{eq}}$$

$$K_{eq} = \frac{2K_1 K_2}{K_1 + K_2}$$

Q. Three rods of material x and three of material y are connected as shown in figure. All the rods are identical in length and cross sectional area. If the end A is maintained at 60°C and the junction E at 10°C , calculate the temperature of the junction B. The thermal conductivity of x is $800 \text{ W Wm}^{-1}\text{C}^{-1}$ and that of y is $400 \text{ Wm}^{-1}\text{C}^{-1}$.

$$R_1 = \frac{l}{K_1 A} \text{ (let)}$$



Sol.
$$\frac{x - 60}{R_1} + \frac{x - 10}{\left(\frac{R_1 + R_2}{2}\right)} = 0$$

$$\frac{x - 60}{\frac{K_1 A}{L_1}} + \frac{2(x - 10)^2}{\frac{K_1 A}{L_1} + \frac{K_2 A}{L_2}} = 0$$

$$K_1 = 800 = 2K$$

$$K_2 = 400 = K$$

$$K_1(x - 60) + \frac{K_1 K_2}{(K_1 + K_2)} (2(x - 10)) = 0$$

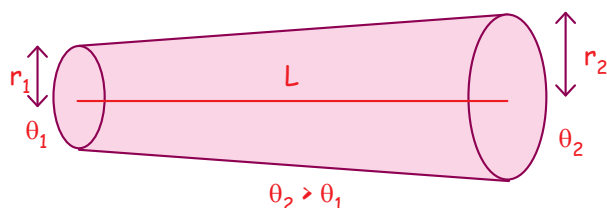
$$x - 60 + 2K_2 \frac{(x - 10)}{K_1 + K_2} = 0$$

$$x - 60 + \frac{2K(x - 10)}{3K} = 0$$

$$3x - 180 + 2x - 20 = 0$$

$$x = 40^\circ\text{C}$$

Q. Find the rate of heat flow through a cross section of the rod shown in figure ($\theta_2 > \theta_1$). Thermal conductivity of the material of the rod is K .

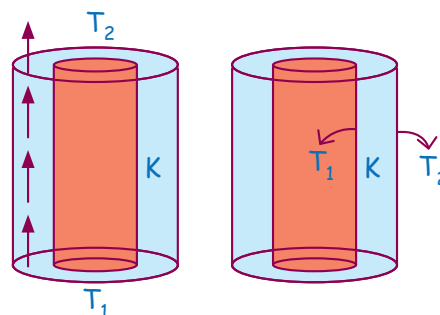


Sol.
$$i = \frac{\theta_2 - \theta_1}{R_{eq}} = \frac{(\theta_2 - \theta_1)}{L} K \pi r_1 r_2$$

Q. A hollow tube has a length l , inner radius R_1 and outer radius R_2 . The material has a thermal conductivity K . Find the heat flowing through the walls of the tube if (a) the flat ends are maintained at temperature T_1 and T_2 ($T_2 > T_1$) (b) the inside of the tube is maintained at temperature T_1 and the outside is maintained at T_2 .

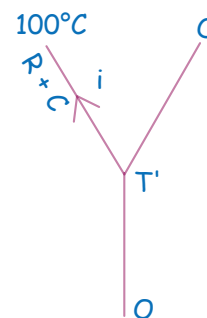
Sol.

$$\frac{l}{K \pi (r_2^2 - r_1^2)}$$



Q. Three rods of Copper, Brass and steel are welded together to form a Y-shaped structure. Area of cross-section of each rod = 4 cm^2 . End of copper rod is maintained at 100°C where as ends of brass and steel are kept at 0°C . Lengths of the copper, brass and steel rods are 46, 13 and 12 cms respectively. The rods are thermally insulated from surrounding except at ends. Thermal conductivities of copper, brass and steel are 0.92, 0.26 and 0.12 CGS units respectively. Rate of heat flow through copper rod is:

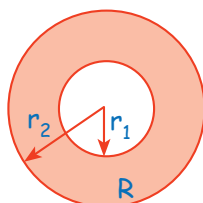
Sol.
$$T' = i = \frac{T' - 100}{R_1}$$



Q. Two thin metallic spherical shells of radii r_1 and r_2 ($r_1 < r_2$) are placed with their centres coinciding. A material of thermal conductivity K is filled in the space between the shells. The inner shell is maintained at temperature θ_1 and the outer shell at temperature θ_2 ($\theta_1 < \theta_2$). The rate at which heat flows radially through the material is:

Sol. Thermal resistance of spherical shell $= \frac{dr}{4\pi K r_1 r_2}$

$$\text{Thermal current } \frac{\Delta T}{R} = \frac{(\theta_2 - \theta_1)}{\left(\frac{r_2 - r_1}{4\pi K r_1 r_2}\right)} = \frac{4\pi K r_1 r_2 (\theta_2 - \theta_1)}{(r_2 - r_1)}$$



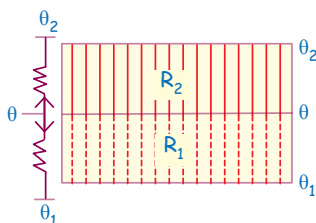
Q. The temperature θ at the junction of two insulating sheets, having thermal resistances R_1 and R_2 as well as top and bottom temperatures θ_1 and θ_2 (as shown in figure) is given by:

Sol. $\frac{\theta - \theta_2}{R_1} + \frac{\theta - \theta_1}{R_2} = 0$

$$\frac{\theta}{R_2} - \frac{\theta_2}{R_2} + \frac{\theta}{R_1} - \frac{\theta_1}{R_1} = 0$$

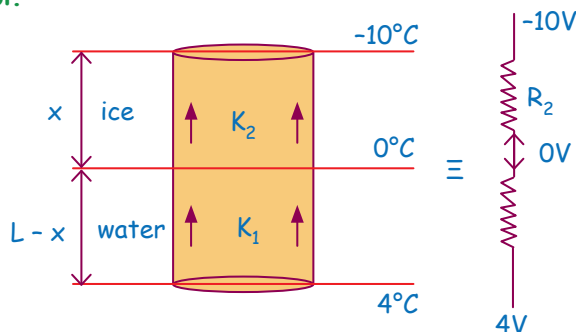
$$\theta \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = \frac{\theta_2}{R_2} + \frac{\theta_1}{R_1}$$

$$\frac{\theta(R_1 + R_2)}{R_1 R_2} = \frac{\theta_2 R_1 + \theta_1 R_2}{R_1 R_2}$$



Q. Consider a lake of depth L the temperature of base is constant at 4°C and atmosphere temperature is -10°C . Thermal conductivity of water is K_1 and for ice is K_2 . Find expression for maximum thickness of ice that can be formed theoretically.

Sol.



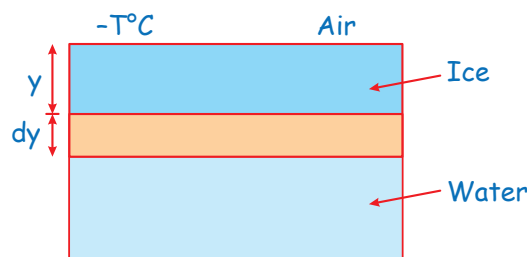
$$\frac{d\theta}{dt} = \frac{K_1 A (4 - 0)}{(L - x)} = \frac{K_2 A (0 - (-10))}{x}$$

Heat Transfer

GROWTH OF ICE ON LAKE

When the temperature of the air is less than 0°C , the cold air near the surface of the pond takes heat (latent) from the water which freezes it and a layer of ice is formed.

Consequently, the thickness of the ice layer keeps increasing with time. Let y be the thickness of ice layer in the lake at any instant t and atmospheric temperature be $-T^\circ\text{C}$. If the thickness is increased by dy in time dt , then the amount of heat flowing through the slab in time dt is given by $dQ_2 = mL = \rho(dt)A)L$ (see figure).



The temperature of water in contact with lower surface of ice will be zero. If A is the area of lake, heat escaping through ice in time dt ,

$$dQ_1 = \frac{KA[0 - (-T)]dt}{y}$$

As $dQ_1 = dQ_2$, hence, rate of growth of ice will be

$$\frac{dy}{dt} = \frac{kT}{\rho Ly}$$

So, the time taken by ice to grow to a thickness y is

$$t = \frac{\rho L}{kT} \int_0^y y dy = \frac{\rho L}{2kT} y^2$$

If the thickness is increased from y_1 to y_2 then time taken,

$$t = \frac{\rho L}{kT} \int_{y_1}^{y_2} y dy = \frac{\rho L}{2kT} (y_2^2 - y_1^2)$$

RADIATION

★ The process of the transfer of heat from one place to another place without heating the intervening medium is called radiation.

★ All objects emit radiations at above absolute zero temp ($T > 0 \text{ KE}$) and all objects absorb some of the radiation that falls on them from other objects.

★ All radiations are electromagnetic waves and their sources are vibrations of charged particles in atoms and molecules.

★ Thermal radiations travels with the speed of light and move in a straight line and they can also travel through vacuum.

Prevost Theory

Prevost Theory of Heat Exchange

All body radiates thermal radiation at all temperature ($T > 0 \text{ K}$)

Amount of rate of heat radiates $\left(\frac{dQ}{dt}\right)$ by body

depends on temperature of body, surface area nature of emitting surface.

A body also absorbs part of the thermal radiation emitted by the surrounding bodies when this radiation falls on it. If a body radiates more than what it absorbs, its temperature falls. If a body radiates less than what it absorbs, its temperature rises. And if the temperature of a body is equal to the temperature of its surroundings it radiates at the same rate as it absorbs.

Stefan Boltzmann Law

Suppose a body of temperature T is placed in surrounding of temperature T_s then

The amount of radiation emitted per unit time $\left(\frac{dQ}{dt}\right)$ is proportional to

$$\frac{d\theta}{dt} \propto T^4$$

$$\propto A$$

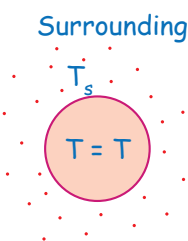
$$\left(\frac{d\theta}{dt}\right) = e\sigma AT^4$$

$e = 1$ for black body

$$\sigma = 5.67 \times 10^{-8} \text{ J/m}^2\text{K}^4 \text{ sec}$$

$T_s, T \rightarrow$ Put in Kelvin

$$\left(\frac{d\theta}{dt}\right)_{\text{radiate}} = eA\sigma T^4$$



$$\left(\frac{d\theta}{dt}\right)_{\text{absorb}} = eA\sigma T_s^4$$

$$\left(\frac{d\theta}{dt}\right)_{\text{net}} = eA\sigma T^4 - eA\sigma T_s^4 \text{ (Let } T > T_s)$$

$$\left(\frac{d\theta}{dt}\right)_{\text{net}} = eA\sigma(T^4 - T_s^4)$$

Dekho bhai, mudde ki baat yeh hai ki agar body ke paas temperature hai, to woh heat radiate karegi, aur agar surroundings ke paas temperature hai, to body heat absorb bhi karegi. If temp of body is greater than temp of surrounding (If $T > T_s$) then body heat radiate jyada karegi aur absorb kam karegi means overall body ka temp kam hota jayega..... bas hame isi par khelna hai ki body kitna radiate karegi, rate of fall of temp kya hoga aur kis time par kitna temp hoga.



$$\left(\frac{d\theta}{dt}\right)_{\text{net}} = eA\sigma(T^4 - T_s^4)$$

or

$$-ms \frac{dT}{dt} = eA\sigma(T^4 - T_s^4)$$

$$-\frac{dT}{dt} = \frac{eA\sigma(T^4 - T_s^4)}{ms} = \text{rate of cooling}$$

$$\left(\frac{d\theta}{dt}\right)_{\text{net}} = eA\sigma(T^4 - T_s^4)$$

$$-\frac{dT}{dt} = \frac{eA\sigma(T^4 - T_s^4)}{ms}$$

Rate of cooling/Rate of fall of temperature

Very Important



Q. A body of emissivity ($e = 0.75$), surface area of 300 cm^2 and temperature 227°C is kept in a room at temperature 27°C . Calculate the initial value of net power emitted by the body.

Sol. $P = e\sigma A (T^4 - T_0^4)$

$$= (0.75) (5.67 \times 10^{-8}) (300 \times 10^{-4})$$

$$\times \{(500)^4 - (300)^4\}$$

$$= 69.4 \text{ Watt.}$$

Ye mat bhulna ki yahan par temp ko Kelvin mein put karna hai.

Q. We have one sphere of radius R and a cube of length R at same temperature made up of same material, same identical surrounding same finishing area.

1. Find ratio of rate of heat loss

Sol.
$$\frac{\left(\frac{d\theta}{dt}\right)_{\text{sphere}}}{\left(\frac{d\theta}{dt}\right)_{\text{cube}}} = \frac{A_1}{A_2} = \frac{4\pi R^2}{6R^2} = \frac{2\pi}{3}$$

2. Find ratio of rate of cooling

Sol.
$$\frac{dT}{dt} = \frac{eA\sigma(T^4 - T_s^4)}{ms}$$

$$\frac{dT}{dt} \propto \frac{A}{m} \propto \frac{A}{\rho(\text{Volume})} \propto \frac{A}{V}$$

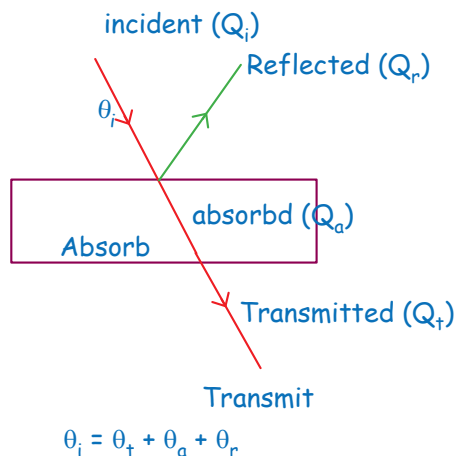
$$\frac{\left(\frac{dT}{dt}\right)_{\text{sphere}}}{\left(\frac{dT}{dt}\right)_{\text{cube}}} = \frac{A_1/V_1}{A_2/V_2} = \frac{4\pi R^2/(4/3\pi R^3)}{6R^2/R^3} = \frac{3}{6} = \frac{1}{2}$$

Q. If rate of heat radiation from a body at temperature 273°C is E . Find rate of heat radiate when body is at 546°C .

Sol.
$$\left(\frac{d\theta}{dt}\right)_{\text{radiate}} = eA\sigma T^4$$

$$\frac{\left(\frac{d\theta}{dt}\right)_1}{\left(\frac{d\theta}{dt}\right)_2} = \frac{T_2^4}{T_1^4} = \frac{(546 + 273)^4}{(273 + 273)^4} = \left(\frac{3}{2}\right)^4 = \frac{81}{16}$$

Radiation Falling on Body



$$I = \frac{\theta_t}{\theta_i} + \frac{\theta_a}{\theta_i} + \frac{\theta_r}{\theta_i}$$

$\swarrow$ Transmissivity (t) $\swarrow$ Absorptivity (a) $\swarrow$ Reflectivity (r)

$$1 = t + a + r \quad (\text{Total radiation conserved})$$

Absorptive Power

In particular absorptive power (absorptivity) of a body can be defined as the fraction of incident radiation that is absorbed by the body.

$$a = \frac{\text{Energy absorbed}}{\text{Energy incident}}$$

As all the radiations incident on a black body are absorbed, $a = 1$ for a black body.

Emissive Power (E)

Energy radiated per unit time per unit area per unit solid angle is known as emissive power.

$$E = \frac{Q}{(\Delta A)(\Delta t)(\Delta \Omega)}$$

(Notice that unlike absorptive power, emissive power is not a dimensionless quantity).

Emissivity

$$e = \frac{\text{Emissive power of a body at temperature } T}{\text{Emissive power of a black body at same temperature } T}$$

$$= \frac{E}{E_0}$$

Emissive Power

Amount of heat energy emitted per sec per unit area by radiating surface

$$E = \frac{\text{Energy emitted}}{\text{Area time}} = e\sigma T^4$$

Absorbing power (a): Ratio of radiant energy absorbed by the body to the total amount of radiant energy incident on it.

$$a = \frac{\text{Energy Absorb}}{\text{Energy incident}}$$

$$\text{Emissivity (e)} = \frac{B(\text{body})}{E(\text{Black body})}$$

= Ratio of emissive power of given body to the emissive power of black body

Kirchoff's law

$$\frac{E(\text{body})}{a(\text{Body})} = \text{Constant} = \frac{E(\text{Black body})}{a(\text{Black body})}$$

$$= E(\text{Black body})$$

$$\frac{e\sigma T^4}{a} = \sigma T^4 \Rightarrow e = a$$

Emissive power $\uparrow \Rightarrow$ absorbing power $\uparrow$

For black body $a = 1, e = 1$

$$a = e$$

Good absorber are good emitter

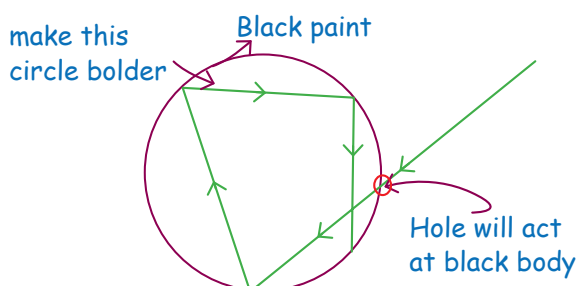
Black body (BB)

A perfectly black body is one which absorbs all the heat radiations of whatever wavelength, incident on it. It neither reflects nor transmits any of the incident radiation and therefore appears black whatever be the colour of the incident radiation.

★ $a = 1, e = 1$

★ Colour black ho koi jaruri nahi hai.

Ex. Sun

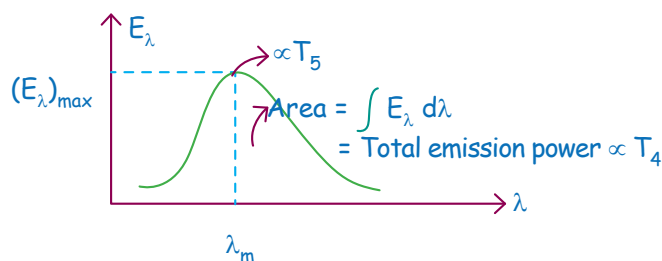


Spectral Emissive Power (E_λ)

Emissive power per unit wavelength range at wavelength λ is known as spectral emissive power, E_λ . If E is the total emissive power and E_λ is spectral emissive power, they are related as follows,

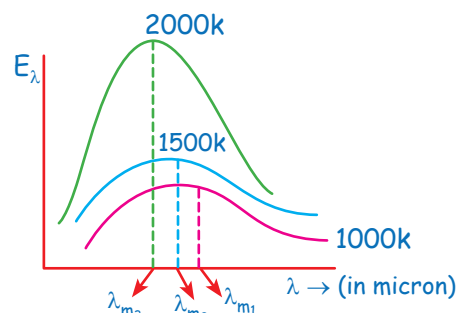
$$E = \int_0^\infty E_\lambda d\lambda \text{ and Emissive Power } (E_\lambda) = \frac{dE}{d\lambda}$$

$$\text{Emission power} = E = \int E_\lambda d\lambda$$



Nature of Thermal Radiations (Wien's Displacement Law)

From the energy distribution curve of black body radiation, the following conclusions can be drawn:



From the study of energy distribution of black body radiation discussed as above, it was established experimentally that the wavelength (λ_m) corresponding to maximum intensity of emission decreases inversely with increase in the temperature of the black body. i.e.

$$\lambda_m \propto \frac{1}{T} \text{ or } \lambda_m T = b$$

$$b = 2.9 \times 10^{-3} \text{ mK}$$

This is called Wien's displacement law.

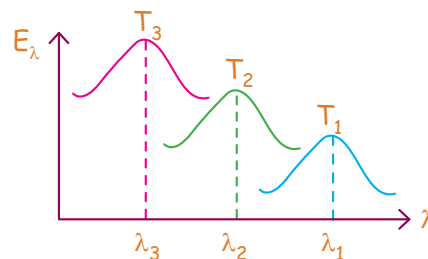
★ If $T \uparrow$, area under curve $\uparrow$ energy emitted $\uparrow$.

★ On increasing the temperature of the graph shifted left (towards shorter wave length), area $\uparrow$.

★ $T \uparrow \lambda_m \downarrow$

★ The energy emitted by the body at different temperatures is not uniform.

Q. Find the relation between T_1, T_2 and T_3



Sol. $\lambda_m \propto \frac{1}{T}$

$$\lambda_3 < \lambda_2 < \lambda_1$$

$$T_3 > T_2 > T_1$$

$$\lambda_m \propto \frac{1}{T}$$

Q. Light from sun is found to have maximum intensity near wavelength of 470 nm. Assuming that surface of sun or black body. Find temperature of sun.

Sol. $\lambda_m = 470 \times 10^{-9} \text{ m}$

$$\lambda_m T = \text{Constant} = 2.9 \times 10^{-3}$$

$$470 \times 10^{-9} \times T = 2.9 \times 10^{-3}$$

$$T = \frac{2.9 \times 10^{-3}}{470 \times 10^{-9}} = 6170 \text{ K}$$

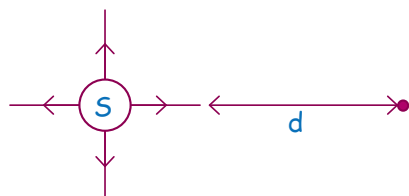
Q. The maximum in the energy distribution spectrum of the sun is at 4753 Å and its temperature is 6050 K. What will be the temperature of the star whose energy distribution shows a maximum at 9506 Å.

Sol. $\lambda_m \propto \frac{1}{T}$

$$\frac{(\lambda_m)_1}{(\lambda_m)_2} = \frac{T_2}{T_1}$$

$$\frac{4753}{9506} = \frac{T_2}{6050}$$

Solar constant: It is defined as the power received per unit area at the Earth's average distance from the Sun.



$$\text{Solar constant} = \frac{\text{Power emitted by Sun}}{\text{Area}}$$

$$= \frac{P}{4\pi d^2} = \frac{\sigma A_s T_s^4}{4\pi d^2} = \frac{\sigma R_s^2 T_s^4}{d^2}$$

Ab main neeche kaafi saare questions, hint aur solution ke saath, attach kar raha hoon. Tumhe poori imaandari se inhe khud se solve karna hai.

Believe me, is type ke questions tumhe exam mein dekhne ko milenge.



Q. Two spherical stars A and B emit blackbody radiation. The radius of A is 400 times that of B and A emits 10^4 times the power emitted from B. The ratio of their wavelength λ_A and λ_B at which the peaks occur in their respective radiation curves is (JEE Adv. 2015)

Sol. $\left(\frac{dq}{dt} \right)_A = \frac{(\text{Area})_A T_A^4}{(\text{Area})_B T_B^4}$

$$= \left(\frac{r_A}{r_B} \right)^2 \left(\frac{T_A}{T_B} \right)^4$$

$$\frac{T_A}{T_B} = \frac{1}{2}$$

$$10^4 = \left(\frac{400r_B}{r_B} \right)^2 \left(\frac{T_A}{T_B} \right)^4$$

$$\left(\frac{T_A}{T_B} \right)^4 = \frac{10000}{400 \times 400} = \frac{1}{16} = \frac{1}{2^4}$$

Q. Find the quantity of heat required to convert 40 g of ice at -20°C into water at 20°C . Given $L_{\text{ice}} = 0.336 \times 10^6 \text{ J/kg}$. Specific heat of ice = 2100 J/kg-K , specific heat of water = 4200 J/kg-K

Sol. 40 gm ice $\longrightarrow$ 40 gm water

- 20°C 20°C

$$\Delta\theta = mS_{\text{ice}} \times 20 + mL_f + mS_w \times 20$$

$$= \left(\frac{40}{1000} \times 2100 \times 20 + \frac{40}{1000} \times 336 \right.$$

$$\left. \times 10^6 + \frac{40}{100} \times 4200 \times 20 \right)$$

Q. An aluminium container of mass 100 gm contains 200 gm of ice at -20°C . Heat is added to the system at the rate of 100 cal/s . Find the temperature of the system after 4 minutes (specific heat of ice = 0.5 and $L = 80 \text{ cal/gm}$, specific heat of Al = $0.2 \text{ cal/gm}^\circ\text{C}$)

Sol. $\begin{array}{|l} -20^\circ\text{C} \\ \text{ice} \\ 200 \text{ gm} \end{array} \quad \begin{array}{l} \Delta\theta = 100 \times 60 \times 4 \\ \Delta\theta = 24000 \text{ cal} \end{array} \longrightarrow T^\circ\text{C water}$

$$24000 = 200 \times \frac{1}{2} \times 20 + 200 \times 80 + 200$$

$$\times 1 \times (T - 0) + 100 \times 0.2 \times (T + 20)$$

you can calculate for $T^\circ\text{C}$

Q. The temperature of 100 gm of water is to be raised from 24°C to 90°C by adding steam to it. Calculate the mass of the steam required for this purpose.

Sol. $100 \times 1 \times (30 - 24) = m \times 540 + m \times 1 \times 10$

$$100 \times 60 = m \times 550$$

$$m = \frac{6600}{550} = \frac{660}{55} = \frac{60}{5} = 12$$

Q. A block of mass 2.5 kg is heated to temperature of 500°C and placed on a large ice block. What is the maximum amount of ice that can melt (approx.). Specific heat for the body = $0.1 \text{ cal/gm}^{\circ}\text{C}$.

Sol. $(25 \times 1000) \times (0.1) \times 500 = m \times 80$

$$m = \frac{125000}{80} = \frac{12500}{8} = \frac{12.5}{8} \text{ kg} = 1.56 \text{ kg}$$

Q. A block of ice with mass m falls into a lake. After impact, a mass of ice $m/5$ melts. Both the block of ice and the lake have a temperature of 0°C . If L represents the heat of fusion, the minimum distance the ice fell before striking the surface is

Sol. $mgh = \left(\frac{m}{5}\right) L_f$

$$h = \frac{L_f}{5g}$$

Q. The specific heat of a metal at low temperatures varies according to $S = aT^3$ where a is a constant and T is the absolute temperature. The heat energy needed to raise unit mass of the metal from $T = 1 \text{ K}$ to $T = 2 \text{ K}$ is:

Sol. $d\theta = mS dT$

$$\int d\theta = \int_1^2 m a T^3 dT$$

$$\Delta\theta = \frac{T^4}{4} \Big|_1^2 = \frac{ma}{4} (16 - 1) = \frac{15a}{4} \quad (m = 1)$$

Q. The density of a material A is 1500 kg/m^3 and that of another material B is 2000 kg/m^3 . It is found that the heat capacity of 8 volumes of A is equal to heat capacity of 12 volumes of B. The ratio of specific heats of A and B will be

Sol.

A	B
$8V_0$	$12V_0$

$$\text{Heat capacity} = mS$$

$$m_1 S_1 = m_2 S_2$$

$$\rho_1 V_1 S_1 = \rho_2 V_2 S_2$$

$$1500 \times 8V_0 \times S_1 = 2000 \times 12V_0 S_2$$

$$\frac{S_1}{S_2} = \frac{12 \times 2000}{1500 \times 8} = \frac{2}{1} \Rightarrow 2 : 1$$

Q. M grams of steam at 100°C is mixed with 200g of ice at its melting point in a thermally insulated container. If it produces liquid water at 40°C [heat of vaporization of water is 540 cal/g and heat of fusion of ice is 80 cal/g], the value of M is _____.

Sol. $M_{\text{ice}} L_f + M_{\text{ice}} (40 - 0) C_w = m_{\text{steam}} L_v + m_{\text{steam}} (100 - 40) C_w$

$$\Rightarrow 200[80 + 40(1)] = m(540 + 60(1))$$

$$\Rightarrow 200(120) = m(600)$$

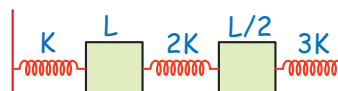
$$m = 40 \text{ gm}$$

Q. A bullet of mass 5g, traveling with a speed of 210 m/s, strikes a fixed wooden target. One half of its kinetic energy is converted into heat in the bullet while the other half is converted into heat in the wood. The rise of temperature of the bullet if the specific heat of its material is $0.030 \text{ cal gm}^{-1}\text{C}^{-1}$ ($1 \text{ cal} = 4.2 \times 10^7 \text{ ergs}$) close to:

Sol. Here $\frac{1}{2} \left(\frac{1}{2} mv^2 \right) = mS\Delta T$

$$\Rightarrow \Delta T = \frac{v^2}{45} = \frac{(210,00)^2}{4 \times 0.03 \times 4.2 \times 10^7} \approx 84.5^{\circ}\text{C}$$

Q. The system shown consists of 3 springs & two rods. If the temperature of the rods is increased by ΔT , calculate the force exerted by springs on wall. Neglect friction and thermal stress & take the coefficient of linear expansion of the material of rods equal to α .



Sol. All the springs will have equal tensions because rods are not rest.

$$T = kx_1 = 2kx_2 = 3kx_3$$

$$x_1 + x_2 + x_3 = \alpha L \Delta T + \alpha \frac{L}{2} \Delta T$$

$$\frac{T}{k} + \frac{T}{2k} + \frac{T}{3k} = \frac{3}{2} \alpha L \Delta T$$

$$\frac{11T}{6k} = \frac{3}{2} \alpha L \Delta T$$

$$T = \frac{9k\alpha L \Delta T}{11}$$

Q. Two separate thin rods of mass M & $2M$ are kept on a smooth horizontal surface such that they are just touching each other. If lengths of the rods are L & $2L$ and coefficients of linear expansion 2α & α , find the displacement of their point of contact when both of them are heated such that their temperature rises by ΔT .



Sol. Let 1 and 2 to be the c.o.m of rod 1 and rod 2

Displacement of 2 w.r.t 1

$$= S_{21} = \frac{L}{2} \times 2\alpha \Delta T + L \times \alpha \Delta T$$

$$S_2 - S_1 = 2L\alpha \Delta T$$

Since there is external force c.o.m. does not move, hence

$$S_1 + 2S_2 = 0$$

$$\Rightarrow -\frac{S_1}{2} = S_2$$

$$\Rightarrow \frac{3S_1}{2} = -2L\alpha \Delta T$$

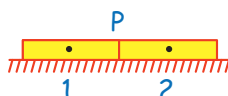
$$S_1 = -\frac{4L\alpha \Delta T}{3}$$

$$S_p = \frac{L}{2} \times 2\alpha \times \Delta T$$

$$S_p - S_1 = L\alpha \Delta T$$

$$S_p = L\alpha \Delta T - \frac{4}{3}L\alpha \Delta T$$

$$S_p = -\frac{\alpha}{3}L\Delta T$$



Q. A rod of length 2m is at a temperature of 20°C . Find the free expansion of the rod, if the temperature is increased to 50°C . Find the thermal stresses produced when the rod is

(i) Fully prevented to expand

(ii) Permitted to expand by 0.4 mm.

$$(Y = 2 \times 10^{11} \text{ Nm}^{-2}; \alpha = 15 \times 10^{-6} \text{ per}^\circ\text{C})$$

Sol. Free expansion of the rod $= \alpha \Delta \theta L$

$$= (15 \times 10^{-6} / ^\circ\text{C}) \times (2\text{m}) \times (50^\circ - 20^\circ) \text{ C}$$

$$= 9 \times 10^{-4} \text{ m} = 0.9 \text{ mm}$$

(i) If the expansion is fully prevented, then strain

$$= \frac{\Delta l}{l}$$

$$\frac{9 \times 10^{-4}}{2} \Rightarrow 4.5 \times 10^{-4}$$

$$\therefore \text{stress} = \text{strain} \times Y = 4.5 \times 10^{-4} \times 2 \times 10^{11}$$

$$= 9 \times 10^7 \text{ Nm}^{-2}$$

(ii) If 0.4 mm expansion is allowed, then length restricted to expand $= 0.9 - 0.4 = 0.5 \text{ mm}$

$$\therefore \text{Strain} = \frac{5 \times 10^{-4}}{2} = 2.5 \times 10^{-4}$$

$$\therefore \text{Thermal stress} = \text{strain} \times Y = 2.5 \times 10^{-4} \times 2 \times 10^{11}$$

$$= 5 \times 10^7 \text{ Nm}^{-2}.$$

Q. Two rods made of different materials are joined end to end and placed between massive walls. The cross-section of the rods are A_1 and A_2 , their lengths L_1 and L_2 , coefficients of linear expansion α_1 and α_2 and the modulus of elasticity of their materials Y_1 and Y_2 respectively. If the rods are heated by $t^\circ\text{C}$. Find the force F which the rods exert on each other.

Sol. Let the first rod expand slightly (say by length δl) and the second rod get compressed by the same amount (since net elongation / compression of the rods is zero).

$\therefore$ Natural increase in length of the first rod (after being heated) when free to expand would have been $\alpha_1 L_1 t$. The expansion allowed is just δl (where $\delta l < \alpha_1 L_1 t$).

$$\therefore \text{Amount of elongation restricted} = \alpha_1 L_1 t - \delta l$$

$$\therefore \text{Strain} = \frac{\text{elongation restricted}}{\text{original length}} = \frac{\alpha_1 L_1 t - \delta l}{L_1 (1 + \alpha_1 t)}$$

$$\text{Since } \alpha_1 t \ll 1$$

$$\therefore 1 + \alpha_1 t \approx 1$$

$$\therefore \text{Strain} = \frac{\alpha_1 L_1 \Delta t - \delta l}{L_1}$$

$$\therefore \text{Stress} = \text{strain} \times Y = \left(\frac{\alpha_1 L_1 \Delta t - \delta l}{L_1} \right) Y_1$$

$$\text{or } F = \text{stress} \times A = \left(\frac{\alpha_1 L_1 \Delta t - \delta l}{L_1} \right) Y_1 A_1$$

$$\text{Similarly, } F = \left(\frac{\alpha_2 L_2 \Delta t + \delta l}{L_2} \right) Y_2 A_2$$

$$\text{or } \delta l = \alpha_1 L_1 \Delta t - \frac{F L_1}{Y_1 A_1} = \frac{F L_2}{Y_2 A_2} - \alpha_2 L_2 \Delta t$$

$$\text{or } F = \frac{(\alpha_1 L_1 + \alpha_2 L_2) \Delta t}{\left(\frac{L_1}{Y_1 A_1} + \frac{L_2}{Y_2 A_2} \right)}$$

- Q.** The specific heat of a substance is given by $C = a + bT$, where $a = 1.12 \text{ kJ kg}^{-1} \text{ } ^\circ\text{C}^{-1}$ and $b = 0.016 \text{ kJ kg}^{-1} \text{ } ^\circ\text{C}^{-2}$. The amount of heat required to raise the temperature of 1.2 kg of the material from 280 K to 312 K is:

Sol. Heat required

$$\begin{aligned} Q &= \int_{280}^{312} m C dT = \int_{280}^{312} (1.2)(a + bT) dT \\ &= 1.2 \left[aT + \frac{bT^2}{2} \right]_{280}^{312} \\ &= 1.2 [1.12(312 - 280) + \frac{0.016}{2} \{(312)^2 - (280)^2\}] \\ &= 1.2 [1.12 \times 32 + \frac{0.016}{2} \times (97344 - 78400)] \\ &= 1.2 [35.84 + 0.008 \times 18944] \\ &= 224.8704 \text{ KJ} = 225 \text{ KJ} \end{aligned}$$

- Q.** A hot black body emits the energy at the rate of $16 \text{ J m}^{-2} \text{ s}^{-1}$ and its most intense radiation corresponds to $20,000 \text{ \AA}$. When the temperature of this body is further increased and its most intense radiation corresponds to $10,000 \text{ \AA}$, then find the value of energy radiated in $\text{J m}^{-2} \text{ s}^{-1}$.

Sol. Wein's displacement law is:

$$\lambda_m \cdot T = b$$

$$\text{i.e. } T \propto \frac{1}{\lambda_m}$$

Here, λ_m becomes half.

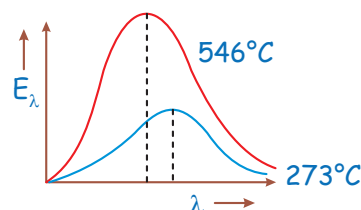
$\therefore$ Temperature doubles.

$$\text{Also } e = \sigma T^4$$

$$\Rightarrow \frac{e_1}{e_2} = \left(\frac{T_1}{T_2} \right)^4$$

$$\Rightarrow e_2 = \left(\frac{T_2}{T_1} \right)^4 \cdot e_1 = (2)^4 \times 16 = 16 \times 16 = 256 \text{ J m}^{-2} \text{ s}^{-1}$$

- Q.** The spectra of a black body at temperatures 273°C and 546°C are shown in the figure. If A_1 and A_2 be the areas under the two curves respectively, the value of $\frac{A_2}{A_1}$ is



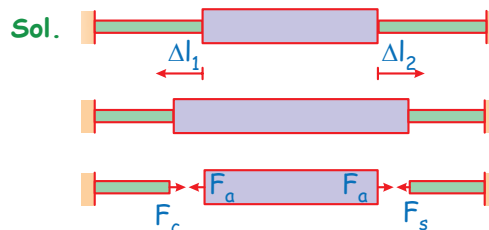
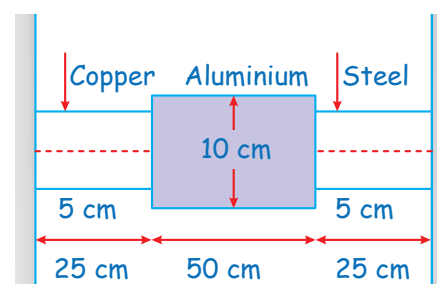
Sol. $\lambda_m T = \text{constant}$

Area under E_λ v/s λ represents total emissive power (E) and $E \propto T^4$

$$\begin{aligned} \text{So, } \frac{A_1}{A_2} &= \frac{E_1}{E_2} = \left(\frac{T_1}{T_2} \right)^4 = \left(\frac{273 + 273}{546 + 273} \right)^4 = \left(\frac{2}{3} \right)^4 = \frac{16}{81} \\ \Rightarrow \frac{A_2}{A_1} &= \frac{81}{16} \end{aligned}$$

- Q.** A composite bar is rigidly attached to the end supports. The temperature of the composite system is raised by 60°C . Find out the stress in three portions of the bar.

$[Y_s = 200 \text{ GPa}, \alpha_s = 12 \times 10^{-6}/^\circ\text{C}, Y_a = 90 \text{ GPa}, \alpha_a = 20 \times 10^{-6}/^\circ\text{C}, Y_c = 100 \text{ GPa}, \alpha_c = 16 \times 10^{-6}/^\circ\text{C}]$



In equilibrium $F_a = F_c$

$$\frac{\Delta l_a - \Delta l_1 - \Delta l_2}{l_a} (Y_a A_a) = \frac{\Delta l_c + \Delta l_1}{l_c} (Y_c A_c)$$

$$\therefore \frac{(50)(20 \times 10^{-6})(60) - \Delta l_1 - \Delta l_2}{50} (90)(10)^2$$

$$= \frac{(25)(16 \times 10^{-6})(60) + \Delta l_1}{25} (100)(5)^2$$

$$\therefore 1.8(6 \times 10^{-2} - \Delta l_1 - \Delta l_2) = (2.4 \times 10^{-2} + \Delta l_1)$$

$$\therefore 2.8\Delta l_1 + 1.8\Delta l_2 = 8.4 \times 10^{-2} \quad \dots(1)$$

Similarly, $F_a = F_s$

$$\therefore \frac{(50)(20 \times 10^{-6})(60) - \Delta l_1 - \Delta l_2}{50} (90)(10)^2$$

$$= \frac{(25)(12 \times 10^{-6})(60) + \Delta l_2}{25} (200)(5)^2$$

$$\therefore 0.9(6 \times 10^{-2} - \Delta l_1 - \Delta l_2) = (1.8 \times 10^{-2} + \Delta l_2)$$

$$0.9\Delta l_1 + 1.9\Delta l_2 = 3.6 \times 10^{-2} \quad \dots(2)$$

Solving Eqs. (1) and (2), we get

$$\Delta l_1 = 2.56 \times 10^{-2} \text{ cm}$$

$$\text{and } \Delta l_2 = 0.68 \times 10^{-2} \text{ cm}$$

Stress in three portions,

$$\sigma_c = \frac{(25)(16 \times 10^{-6})(60) + 2.56 \times 10^{-2}}{25} \times (100 \times 10^9) \text{ N/m}^2$$

$$= 19.84 \times 10^7 \text{ N/m}^2$$

$$\sigma_a = \frac{50 \times (20 \times 10^{-6})(60 - 2.56 \times 10^{-2}) - 0.68 \times 10^{-2}(90 \times 10^9)}{50}$$

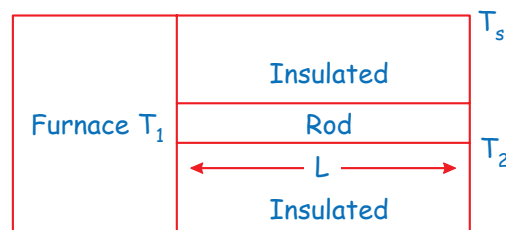
$$= 4.97 \times 10^7 \text{ N/m}^2 \text{ (compressive)}$$

$$\text{and } \sigma_s = \frac{(25)(12 \times 10^{-6})(60) + 0.68 \times 10^{-2}}{25} \times (200 \times 10^9)$$

$$= 19.84 \times 10^8 \text{ N/m}^2 \text{ (compressive)}$$

- Q.** One end of a rod of length L and cross-sectional area A is kept in a furnace of temperature T_1 . The other end of the rod is kept at a temperature T_2 . The thermal conductivity of the material of the rod is K and emissivity of the rod is e . It is given that $T_2 = T_s + \Delta T$, where $\Delta T \ll T_s$, T_s being the temperature of the surroundings. If $\Delta T \propto (T_1 - T_s)$, find the proportionality constant.

Consider that heat is lost only by radiation at the end.



Sol. Rate of heat conduction through rod = rate of heat lost from the right end of the rod.

$$\therefore \frac{KA(T_1 - T_2)}{L} = eA\sigma(T_2^4 - T_s^4) \quad \dots (i)$$

Given that $T_2 = T_s + \Delta T$

$$\therefore T_2^4 = (T_s + \Delta T)^4 = T_s^4 \left(1 + \frac{\Delta T}{T_s}\right)^4$$

Using binomial expansion we have,

$$T_2^4 \approx T_s^4 \left(1 + 4\frac{\Delta T}{T_s}\right) \text{ as } \Delta T \ll T_s$$

$$\therefore T_2^4 - T_s^4 = 4(\Delta T)(T_s^3)$$

Substituting in equation (i) we get

$$\frac{K(T_1 - T_s - \Delta T)}{L} = 4e\sigma T_s^3 \cdot \Delta T$$

$$\text{or } \frac{K(T_1 - T_s)}{L} = \left(4e\sigma T_s^3 + \frac{K}{L}\right) \cdot \Delta T$$

$$\therefore \Delta T = \frac{K(T_1 - T_s)}{(4e\sigma L T_s^3 + K)}$$

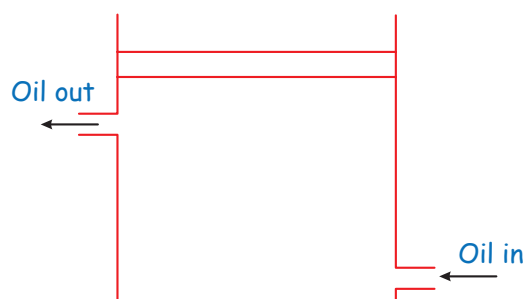
Comparing with the given relation, proportionality constant

$$= \frac{K}{4e\sigma L T_s^3 + K}$$

- Q.** The top of an insulated cylindrical container is covered by a disc having emissivity 0.6 and conductivity 0.167 W/Km and thickness 1cm. The temperature is maintained by circulating oil as shown.

- (a) Find the radiation loss to the surroundings in $\text{J/m}^2\text{s}$ if temperature of the upper surface of disc is 127°C and temperature of surroundings is 27°C .
- (b) Also find the temperature of the circulating oil. Neglect the heat loss due to convection.

$$\sigma = \frac{17}{3} \times 10^{-8} \text{ Wm}^{-2} \text{ K}^{-4}$$



Sol. (a) Rate of heat loss per unit area due to radiation

$$I = e\sigma(T^4 - T_0^4)$$

$$\text{Here } T = 127 + 273 = 400 \text{ K}$$

$$\text{and } T_0 = 27 + 273 = 300 \text{ K}$$

$$\therefore I = 0.6 \times \frac{17}{3} \times 10^{-8} [(400)^4 - (300)^4] = 595 \text{ watt/m}^2$$

(b) Let θ be the temperature of the oil. Then rate of heat flow through conduction = rate of heat loss due to radiation

$$\therefore \frac{\text{Temperature difference}}{\text{Thermal resistance}} = (595)A$$

$$\therefore \frac{(\theta - 127)}{\frac{l}{kA}} = (595)A$$

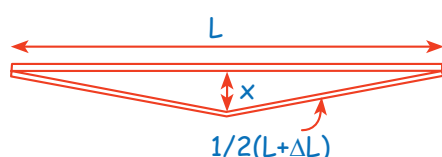
Here A = area of disc; k = thermal conductivity and l = thickness (or length) of disc

$$\therefore (\theta - 127) \frac{k}{l} = 595$$

$$\therefore \theta = 595 \frac{l}{k} + 127$$

$$= \frac{595 \times 10^{-2}}{0.167} + 127 = 162.6^\circ\text{C}$$

Q. A rail track made of steel having length 10 m is clamped on a railway line at its two ends (Fig.). On a summer day due to rise in temperature by 20°C , it is deformed as shown in figure. Find x (displacement of the centre in cm) if $\alpha_{\text{steel}} = 1.2 \times 10^{-5} / ^\circ\text{C}$. Roundoff to nearest integer.



$$\text{Sol. } x = \sqrt{\left(\frac{L}{2} + \frac{\Delta L}{2}\right)^2 - \left(\frac{L}{2}\right)^2} = \frac{1}{2}\sqrt{2L\Delta L}$$

$$\Delta L = \alpha L \Delta T$$

$$\therefore x = \frac{L}{2}\sqrt{2\alpha\Delta T} \approx 0.11\text{m} = 11 \text{ cm}$$

Q. Three rods of equal length are joined to form an equilateral $\triangle ABC$. D is the midpoint of AB. The coefficient of linear expansion is α_1 for AB, α_2 for AC and BC. If the angle $\angle ABC$ remains constant for small changes in temperature then

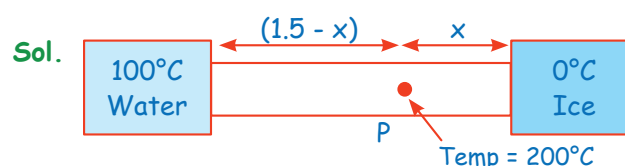
calculate $\frac{\alpha_1}{\alpha_2}$.

$$\text{Sol. } \cos 60^\circ = \frac{l_1^2 + l_2^2 - l_2^2}{2l_1l_2} = \frac{l_1}{2l_2}$$

$$\frac{1}{2} = \frac{l(1 + \alpha_1\Delta\theta)}{2l(1 + \alpha_2\Delta\theta)}$$

$$\alpha_1 = \alpha_2 \Rightarrow \frac{\alpha_1}{\alpha_2} = 1$$

Q. One end of copper rod of uniform cross-section and of length 1.5 meters is in contact with melting ice and the other end with boiling water. At what point along its length should a temperature of 200°C be maintained, so that in steady state, the mass of ice melting is equal to that of steam produced in the same interval of time? Assume that the whole system is insulated from the surroundings.



Sol.

$$\frac{kA}{x}(200^\circ - 0^\circ) = 80 \left(\frac{dm}{dt} \right)_{\text{ice}} \quad \dots(i)$$

$$\frac{kA}{(1.5 - x)}(200^\circ - 100^\circ) = 540 \left(\frac{dm}{dt} \right)_{\text{vapour}} \quad \dots(ii)$$

$$\left(\frac{dm}{dt} \right)_{\text{ice}} = \left(\frac{dm}{dt} \right)_{\text{vapour}}$$

Q. A solid receives heat by radiation over its surface at the rate of 4 kW. The heat convection rate from the surface of solid to the surrounding is 5.2 kW, and heat is generated at a rate of 1.7 kW over the volume of the solid. The rate of change of the average temperature of the solid is 0.5°C s^{-1} . Find the heat capacity of the solid.

Sol. $\frac{d\theta}{dt} = ms \frac{dT}{dt}$

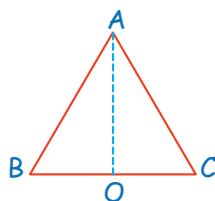
ms = Heat capacity

$$\frac{d\theta}{dt} = (4 + 1.7 - 5.2) \text{ kw}$$

$$\frac{d\theta}{dt} = 0.5^\circ \text{Cs}^{-1}$$

$$1000 \text{ J } (^\circ\text{C})^{-1}$$

- Q.** Three aluminium rods of equal length form an equilateral triangle ABC. Taking O (mid point of rod BC) as the origin. Find the increase in Y-coordinate of center of mass per unit change in temperature of the system. Assume the length of the each rod is 2 m, and $\alpha_{al} = 4\sqrt{3} \times 10^{-6} / ^\circ\text{C}$.



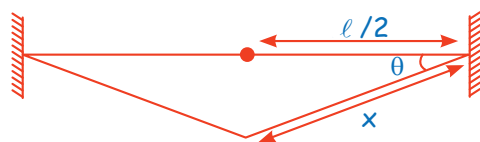
Sol. $\Delta y = \frac{1}{3} \left(\frac{\sqrt{3}}{2} \ell \right) \alpha \Delta \theta$

$$\frac{\Delta y}{\Delta \theta} = \frac{1}{3} \left(\frac{\sqrt{3}}{2} \ell \right) \alpha$$

$$4 \times 10^{-6} \text{m}/^\circ\text{C}$$

- Q.** A wire of cross-sectional area $4 \times 10^{-4} \text{ m}^2$, modulus of elasticity $2 \times 10^{11} \text{ N/m}^2$ and length 1 m is stretched between two vertical rigid poles. A mass of 1 kg is suspended at its middle. Calculate the angle it makes with the horizontal.

Sol. $2T \sin \theta = mg$



$$T = \frac{mg}{2 \sin \theta} \quad \dots(i)$$

$$\frac{T}{A} = Y \frac{\Delta \ell}{\ell} \quad \dots(ii)$$

$$\cos \theta = \frac{\ell/2}{x} \Rightarrow x = \frac{\ell}{2} \sec \theta \quad \dots(iii)$$

$$\Delta \ell = 2x - \ell$$

$$1/200 \text{ rad}$$

- Q.** A solid sphere of radius R made of material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area A floats on the surface of the liquid. When a mass m is placed on the piston to compress the liquid, the fractional change in the radius of the sphere $\delta R/R$ is

Sol. $V = \frac{4}{3} \pi R^3$

$$\frac{\Delta V}{V} = 3 \frac{\Delta R}{R}$$

$$B = K = \frac{P}{\Delta V / V} \Rightarrow \frac{\Delta V}{V} = \frac{mg}{AK} = \frac{3\Delta R}{R}$$

- Q.** Two bodies A and B have thermal emissivities of 0.01 and 0.81 respectively. The outer surface areas of the two bodies are the same. The two bodies radiate energy at the same rate. The wavelength λ_B , corresponding to the maximum spectral radiance in the radiation from B, is shifted from the wavelength corresponding to the maximum spectral radiance in the radiation from A by $1.00 \mu\text{m}$. If the temperature of A is 5802 K,

Sol. $e_A = 0.01, e_B = 0.81$

Area = same for both

$$\frac{dQ}{dt} = e\sigma AT^4 = \text{same for both}$$

$$\sigma e_A T_A^4 = \text{same for both}$$

$$\left(\frac{T_A}{T_B} \right)^4 = \frac{e_B}{e_A} = 81 = 3^4$$

$$\frac{T_A}{T_B} = 3 \Rightarrow T_B = \frac{T_A}{3} = 1934$$

also

$$\lambda_B T_B = \lambda_A T_A$$

$$\lambda_B \times 1934 = (\lambda_A - 1)5802$$

$$\lambda_B = 3\lambda_B - 3$$

$$2\lambda_B = 3$$

$$\lambda_B = 1.5 \mu\text{m}$$

- Q.** An isosceles triangle is formed with a rod of length l_1 and coefficient of linear expansion α_1 for the base and two thin rods each of length l_2 and coefficient of linear expansion α_2 for the two pieces, if the distance between the apex and the midpoint of the base remain unchanged as

the temperatures varied show that $\frac{l_1}{l_2} = 2 \sqrt{\frac{\alpha_2}{\alpha_1}}$.

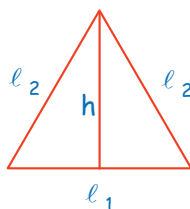
Sol. $\ell_2^2 = h^2 + \frac{\ell_1^2}{4}$

$2\ell_2 d\ell_2 = \frac{\ell_1^2}{2} d\ell_1 \quad \dots(i)$

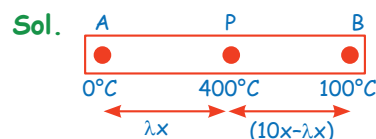
Put $d\ell_2 = \ell_2 \alpha_2 \Delta\theta$

$d\ell_1 = \ell_1 \alpha_1 \Delta\theta$

in equation (i)



Q. A metal rod AB of length $10x$ has its one end A in ice at 0°C and the other end B in water at 100°C . If a point P on the rod is maintained at 400°C , then it is found that equal amounts of water and ice evaporate and melt per unit time. The latent heat of evaporation of water is 540 cal/g and latent heat of melting of ice is 80 cal/g . If the point P is at a distance of λx from the ice end A, find the value of λ . [Neglect any heat loss to the surrounding]



For ice melting at 0°C

$L_1 \frac{dm_1}{dt} = \frac{kA400}{\lambda x} \quad \dots(i)$

For water evaporation at 100°C

$L_2 \frac{dm_2}{dt} = \frac{kA(400 - 100)}{(10 - \lambda)x}$

$\frac{dm_1}{dt} = \frac{dm_2}{dt}$

NEWTON'S LAW OF COOLING

Waise to yeh sirf Advance mein hain, Mains mein nahi. Lekin I will suggest ki aap isse zaroor padhein, kyunki Mains wale bahut kamine hain —har saal ek sawaal pooch hi lete hain.

For small temperature difference between a body and its surrounding, the rate of cooling of the body is directly proportional to the temperature difference and the surface area exposed.

$\frac{dT}{dt} \propto (T - T_s),$

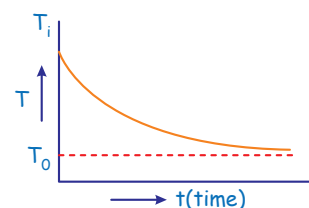
where T and T_s are temperature corresponding to object and surroundings.

From above expression, $\frac{dT}{dt} = -k(T - T_0) \quad \dots(i)$

This expression represents Newton's law of cooling. It can be derived directly from Stefan's law, when cooling is occur only from radiations.

$k = \frac{4e\sigma T_0^3}{mc} A$

Now $\frac{dT}{dt} = -k [T - T_0] \Rightarrow \int_{T_i}^{T_f} \frac{dT}{(T - T_0)} = \int_0^t -k dt$



where T_i = initial temperature of object and

T_f = final temperature of object.

$\Rightarrow \ln \frac{(T_f - T_0)}{(T_i - T_0)} = -kt \Rightarrow (T_f - T_0) = (T_i - T_0) e^{-kt}$

$\Rightarrow T_f = T_0 + (T_i - T_0) e^{-kt} \quad \dots(ii)$

Approximate Method for Applying Newton's Law of Cooling

Sometime when we need only approximate values from Newton's law, we can assume a constant rate of cooling, which is equal to the rate of cooling corresponding to the average temperature of the body during the interval.

$\left(\frac{dT}{dt}\right) = -k(T_{\text{avg}} - T_0)$

$\frac{T_f - T_i}{t} = -k \left(\frac{T_i + T_f}{2} - T_0 \right) \quad \dots(iii)$

If T_i & T_f be initial and final temperature of the body then,

$T_{\text{avg}} = \frac{T_i + T_f}{2}$

Remember equation (iii) is only an approximation and equation (ii) must be used for exact values.

Q. A beaker of boiled water cools from 80°C to 40°C in 6 minutes. What is the time taken to cool from 40°C to 30°C . The temperature of the surrounding is 20°C .

Sol. $t_0 = \frac{2.3026}{K} \log_{10} \frac{T_1 - T_0}{T_2 - T_0}$

$$t_0 = \frac{2.3026}{K} \log_{10} \frac{80 - 20}{40 - 20} \Rightarrow t_0 = \frac{2.3026}{K} \log_{10} \frac{60}{20}$$

$$6 = \frac{2.303}{K} \log_{10} (3) \quad \dots (i)$$

$$t = \frac{2.303}{K} \log_{10} \frac{40 - 20}{30 - 20} \Rightarrow t = \frac{2.303}{K} \log_{10} \frac{20}{10}$$

$$t = \frac{2.303}{K} \log_{10} (2) \quad \dots (ii)$$

From eq (i) & (ii) we get

$$\frac{6}{t} = \frac{\log_{10} (3)}{\log_{10} (2)}$$

$$t = 6 \times 0.631 = 3.79 \text{ minutes}$$

Q. A body takes 4 minutes to cool from 100°C to 70°C . If temperature of the surrounding is 20°C . The time taken by it to cool from 70°C to 50°C is

Sol. $\frac{dT}{t} = -K \left(\frac{T_1 + T_2}{2} - T_0 \right) \Rightarrow \frac{30}{4} = -K \left(\frac{170}{2} - 20 \right) \quad \dots (i)$

$$\frac{20}{t} = -K \left(\frac{120}{2} - 20 \right) \quad \dots (ii)$$

Divide (i) & (ii) we get

$$\frac{t}{20} \times \frac{30}{4} = \frac{-K(85 - 20)}{-K(60 - 20)} \Rightarrow t = \frac{13}{3} \text{ min}$$

Q. A system S receives heat continuously from an electrical heater of power 10 W. The temperature of S becomes constant at 50°C when the surrounding temperature is 20°C . After the heater is switched off, S cools from 35.1°C to 34.9°C in 1 minute. The heat capacity of S is

Sol. The correct option is c $1500 \text{ J}/^\circ\text{C}$

Formula used : $\frac{dQ}{dt} = k\Delta\theta$, $Q = \frac{dQ}{dt} \times t$, $Q = C\Delta\theta$

Given: $\frac{dQ}{dt} = 10 \text{ W}$ of electric heater, $\Delta\theta = 30^\circ\text{C}$, After heater is switched off it takes 1 min from 35.1°C to 34.9°C .

At 50°C ,

$$\frac{dQ}{dt} = k(50 - 20)$$

$$10 = k(50 - 20)$$

$$\therefore k = \frac{1}{3}$$

At an average temperature of 35°C ,

$$\frac{dQ}{dt} = \frac{1}{3}(35 - 20) = 5 \text{ J/s}$$

$$\text{Heat lost in 1 minute} = \frac{dQ}{dt} \times 60 = 300 \text{ J}$$

Fall in temperature is $0.2^\circ\text{C} = \Delta\theta$

$$\therefore C = \frac{Q}{\Delta\theta} = \frac{300}{0.2} = 1500 \text{ J}/^\circ\text{C}$$